# Optimal Reconfiguration of Systems in Aeronautics

## Lecture 14: "B747 Application"

Bogdan D. Ciubotaru

2024-25

# Outline


1. B747 Progressive Fault Accommodation
2. B747 Admissible and Optimal Model Matching

# Objective I

This lecture provides the full application of fault tolerant control methods developed within these series to the longitudinal motion of the Boeing 747 aircraft.

It shows how these methods behave when applied for the control of the longitudinal motion.

The lecture is organized as follows: the first section presents the results obtained when the linear quadratic regulator is used for the progressive accommodation technique, while the second section presents the results obtained when the eigenvalue assignment design is used in the admissible and optimal model matching technique.

# Outline


1. B747 Progressive Fault Accommodation
   - Flight Condition
   - Nominal System
   - Faulty System
2. B747 Admissible and Optimal Model Matching

# Flight Condition I

Consider a Class III large transport Boeing 747 aircraft in Class I cruise flight.

Let the Category B flight phase be, i.e., gradual maneuvering without precision tracking.

Assume a straight and constant level flight at $40000$ *ft* fixed altitude and $0.8$ Mach; the corresponding steady-state speed is $774$ *fps*. (The CoG position is $(0.25, 0, 0)$ m.a.c. and the aircraft mass is $19792$ *slug*.)

# Open-Loop Model I

The aircraft longitudinal dynamics is described by the state vector

$$x \equiv x_{LON} = \begin{bmatrix} u & w & q & \theta \end{bmatrix}^\top ,$$

where $u$ $(ft/sec)$ and $w$ $(ft/sec)$ represent the inertial velocities in the $x$- and $z$-directions of $F_B$ reference frame; also, $q$ $(rad/sec)$ and $\theta$ $(rad)$ represent the pitch rate and pitch angle, respectively ($F_B$ denotes body-axis reference).

The control input $\delta_E$ $(rad)$ is the elevator deflection.

The linearized model of the system is given by the nominal pair of system and control matrices $(A_n, B_n)$ as

$$A_n = \begin{bmatrix} -0.0069 & 0.0139 & 0.0 & -32.2000 \\ -0.0904 & -0.3147 & 773.9766 & 0.0 \\ 0.0001 & -0.0010 & -0.4284 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \end{bmatrix} \text{ and } B_n = \begin{bmatrix} -0.0002 \\ -18.0610 \\ -1.1577 \\ 0.0 \end{bmatrix},$$

where $A_n \equiv A_{LON}$ and $B_n \equiv B_{LON}$, respectively.

# Closed-Loop System I

To design a linear quadratic regulator, choose the symmetric weighting matrices $Q_n = 10^{-8} \times I_4$ and $R_n = 10^{-4}$, for when full-state feedback design is assumed.

The nominal control matrix

$$K_n = \begin{bmatrix} 0.0076 & -0.0047 & -2.9529 & -3.0015 \end{bmatrix}$$

provides the closed-loop system $\hat{M}_n$, with the nominal closed-loop matrix $\hat{A}_n = A_n - B_n K_n$ as

$$\hat{A}_n = \begin{bmatrix} -0.0069 & 0.0139 & -0.0006 & -32.2006 \\ 0.0475 & -0.3992 & 720.6442 & -54.2093 \\ 0.0089 & -0.0064 & -3.8470 & -3.4748 \\ 0.0 & 0.0 & 1.0 & 0.0 \end{bmatrix},$$

which has the stable spectrum

$$\Lambda(\hat{A}_n) = \{-2.0578 \pm 2.2020\jmath, -0.0687 \pm 0.0907\jmath\} .$$

The nominal operating modes, namely the short-period (SP) $\lambda_{n_{SP}}^{1,2} = -2.0578 \pm 2.2020\jmath$, characterized by the dynamical parameters $(\zeta_n^{SP}, \omega_n^{SP}) = (0.6828, 3.0138)$, respectively the long-period (phugoid (PH)) $\lambda_{n_{PH}}^{1,2} = -0.0687 \pm 0.0907\jmath$, with $(\zeta_n^{PH}, \omega_n^{PH}) = (0.6064, 0.1134)$, are in the area of Level 1 handling qualities for the aircraft longitudinal motion.

# Considered Faults I

The considered faults are the actuator fault defined by the loss-of-effectiveness with $50\,\%$ in the elevator deflection, that can be modeled in the post-fault system by multiplying the control matrix with a factor $\tau_f = 0.5$, i.e., $B_f = (1 - \tau_f) B_n$, respectively the structural fault modeled in the system matrix as $A_f = A_n + \Delta A_f \times I_4$, with $\Delta A_f = \sigma_n^{\max} \times 10^{-3}$, where $\sigma_n^{\max} = 773.9774$ is the maximum singular value of $A_n$.

# Classical and Progressive Accommodation I

In what follows, the numerical computations for the accommodation of the short-period mode dynamical parameters are presented; moreover, though what interests more after the fault is the short-period mode behavior, for completeness, the analysis of the phugoid mode behavior is also given below.

# Classical and Progressive Accommodation II

Next, the accommodation of the short-period mode dynamical parameters using the Newton-Kleinman (NK) algorithm is given in the following table (the instant $k = -1$ indicates the dynamic condition in which the system was before the fault).

| $k$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|
| $\zeta_{f_{SP}}^{NK}$ | 0.2141 | 1.0000 | 1.0000 | 0.6690 |
| $\omega_{f_{SP}}^{NK}$ | 1.9333 | 7.6213 | 3.8046 | 1.9489 |

| $k$ | $3$ | $4$ | $5$ | $6$ | $7$ |
|---|---|---|---|---|---|
| $\zeta_{f_{SP}}^{NK}$ | 0.6597 | 0.6584 | 0.6586 | 0.6586 | 0.6586 |
| $\omega_{f_{SP}}^{NK}$ | 2.1055 | 2.1309 | 2.1320 | 2.1320 | 2.1320 |

# Classical and Progressive Accommodation III

Moreover, the accommodation of the short-period mode dynamical parameters using the Matrix-Sign-Function (MSF) technique is given in the following table.

| $k$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|
| $\zeta_{f_{SP}}^{MSF}$ | 0.2141 | 1.0000 | 0.4268 | 0.9758 |
| $\omega_{f_{SP}}^{MSF}$ | 1.9333 | 49.0624 | 1.1797 | 1.3191 |

| $k$ | $3$ | $4$ | $5$ | $6$ |
|---|---|---|---|---|
| $\zeta_{f_{SP}}^{MSF}$ | 0.6526 | 0.6586 | 0.6586 | 0.6586 |
| $\omega_{f_{SP}}^{MSF}$ | 2.1182 | 2.1320 | 2.1320 | 2.1320 |

# Classical and Progressive Accommodation IV

In this respect, the accommodation of the phugoid mode damping coefficient using the Newton-Kleinman algorithm is given in the following table.

| $k$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|
| $\zeta_{f_{PH}}^{NK}$ | $-0.9920$ | $1.0000$ | $1.0000$ | $1.0000$ |

| $k$ | $3$ | $4$ | $5$ | $6$ | $7$ |
|---|---|---|---|---|---|
| $\zeta_{f_{PH}}^{NK}$ | $1.0000$ | $0.9971$ | $0.9969$ | $0.9969$ | $0.9969$ |

# Classical and Progressive Accommodation V

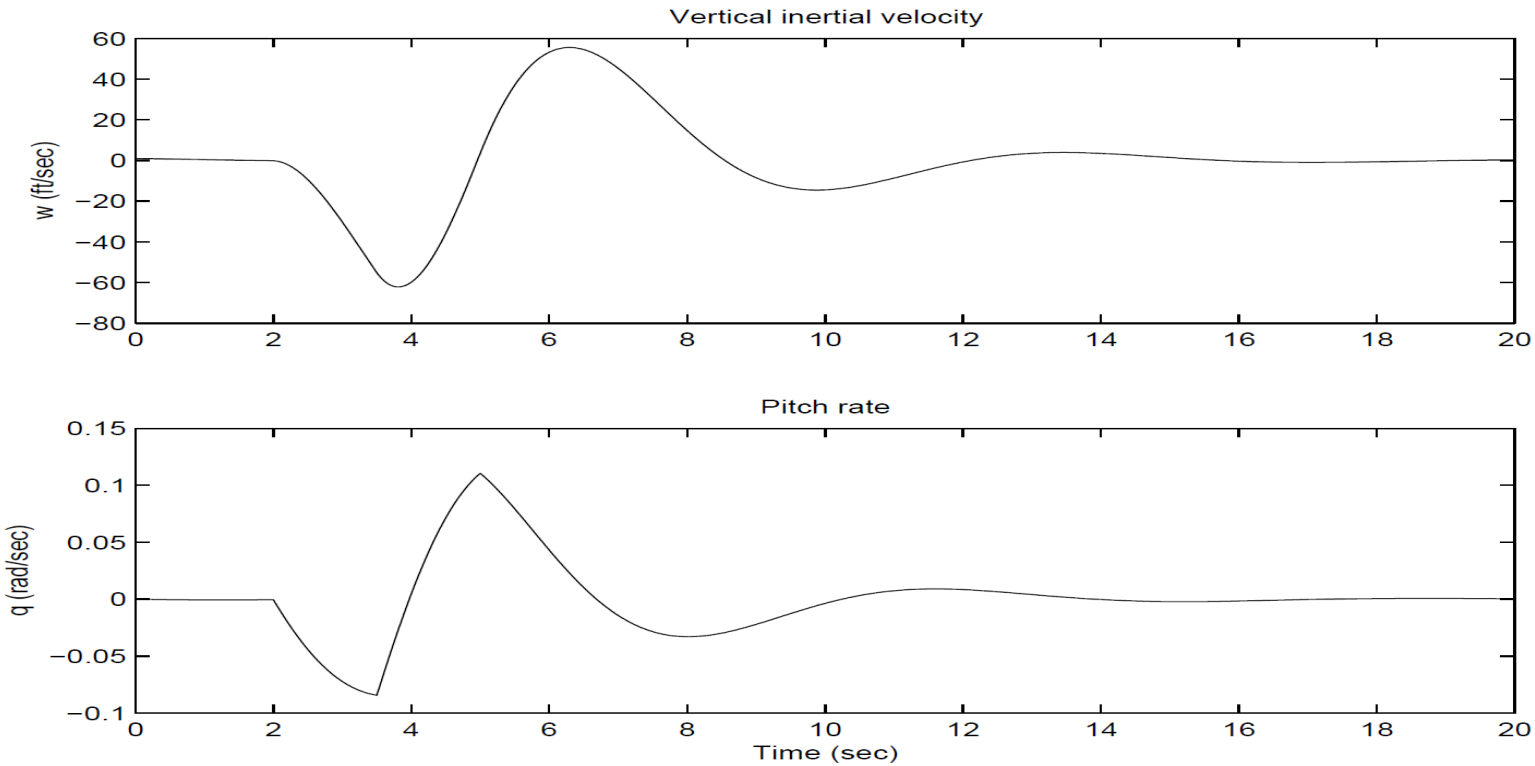
Moreover, again, the accommodation of the phugoid mode damping coefficient using the Matrix-Sign-Function technique is given in the following table.

| $k$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|
| $\zeta_{f_{PH}}^{MSF}$ | $-0.9920$ | $1.0000$ | $1.0000$ | $0.9767$ |

| $k$ | $3$ | $4$ | $5$ | $6$ |
|---|---|---|---|---|
| $\zeta_{f_{PH}}^{MSF}$ | $0.9924$ | $0.9969$ | $0.9969$ | $0.9969$ |

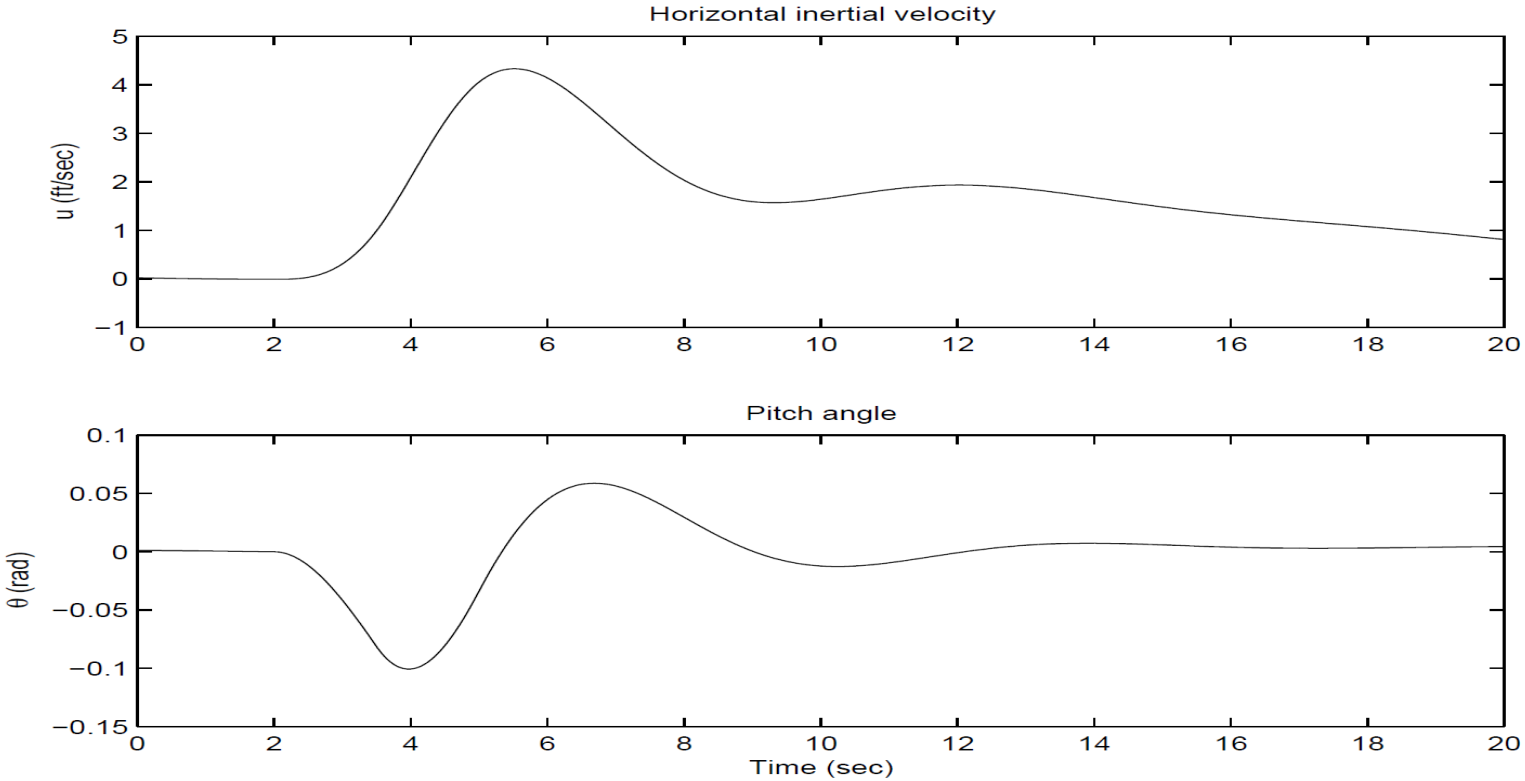
# Classical and Progressive Accommodation VI

In the following figures, the open-loop behavior is shown, respectively for the short-period mode and for the phugoid mode.




Open-Loop Behavior for Longitudinal B747-SP

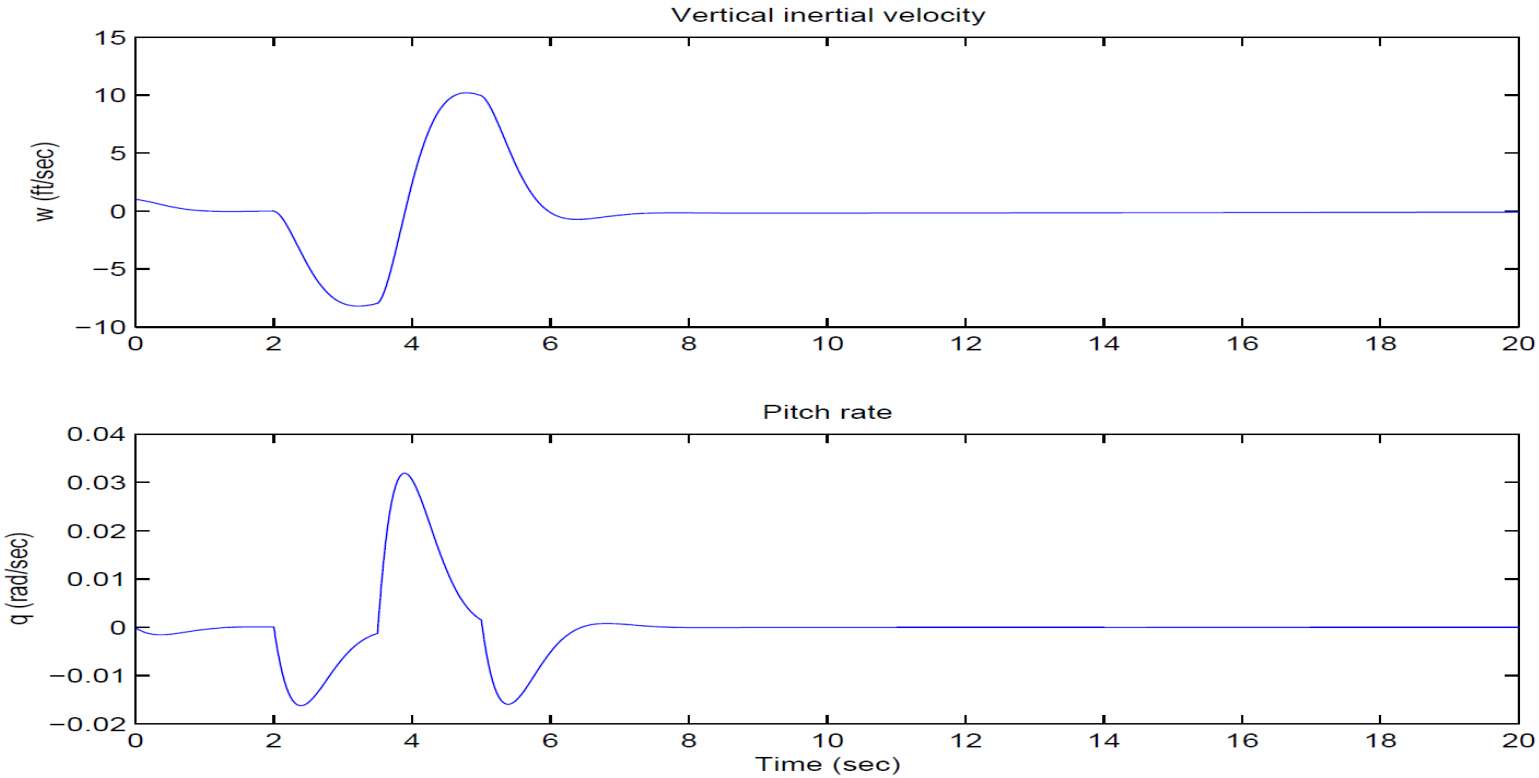
# Classical and Progressive Accommodation VII




Open-Loop Behavior for Longitudinal B747-PH

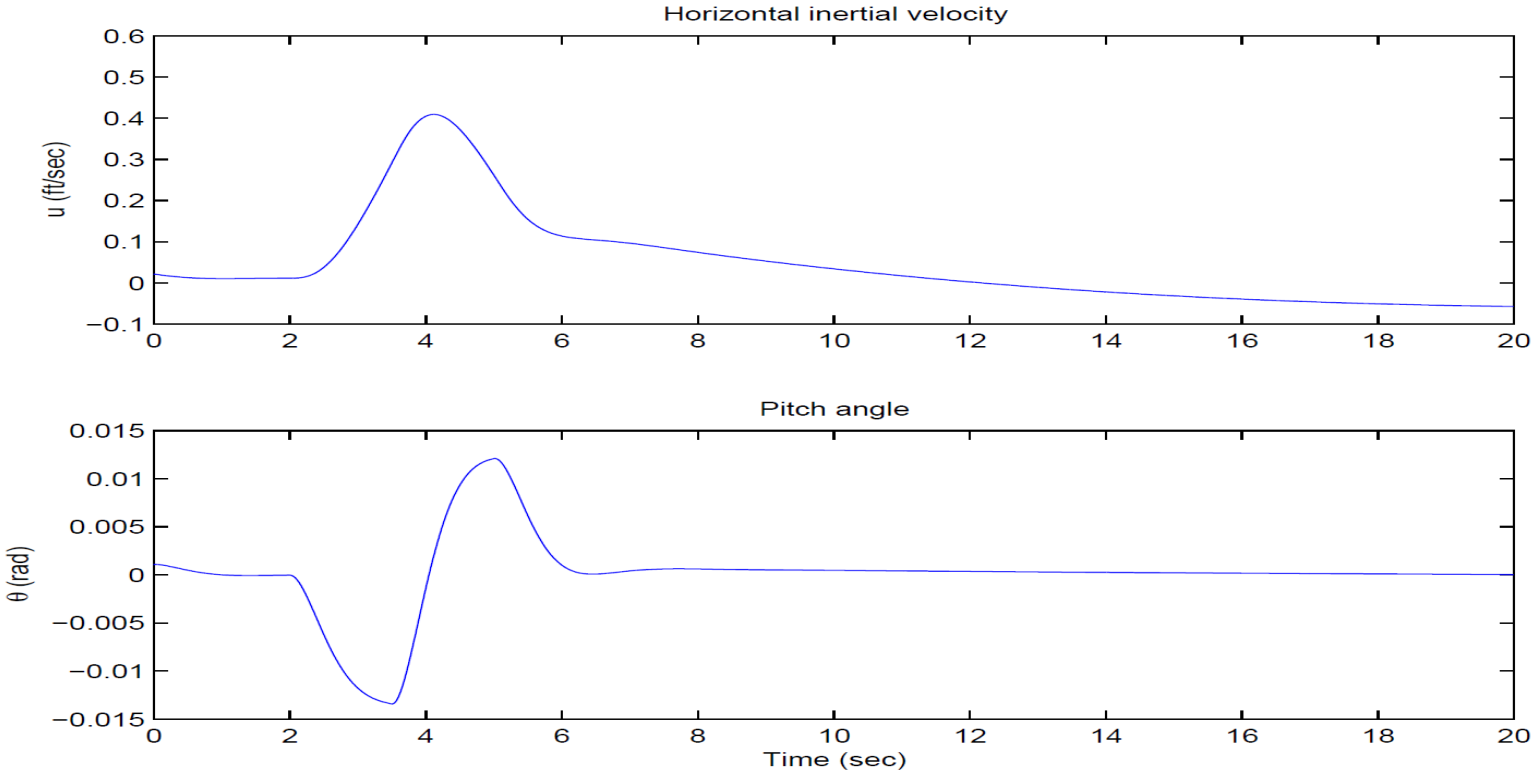
# Classical and Progressive Accommodation VIII

Moreover, in the following figures, the closed-loop behavior is shown, respectively for the short-period mode and for the phugoid mode.




Closed-Loop Behavior for Longitudinal B747-SP

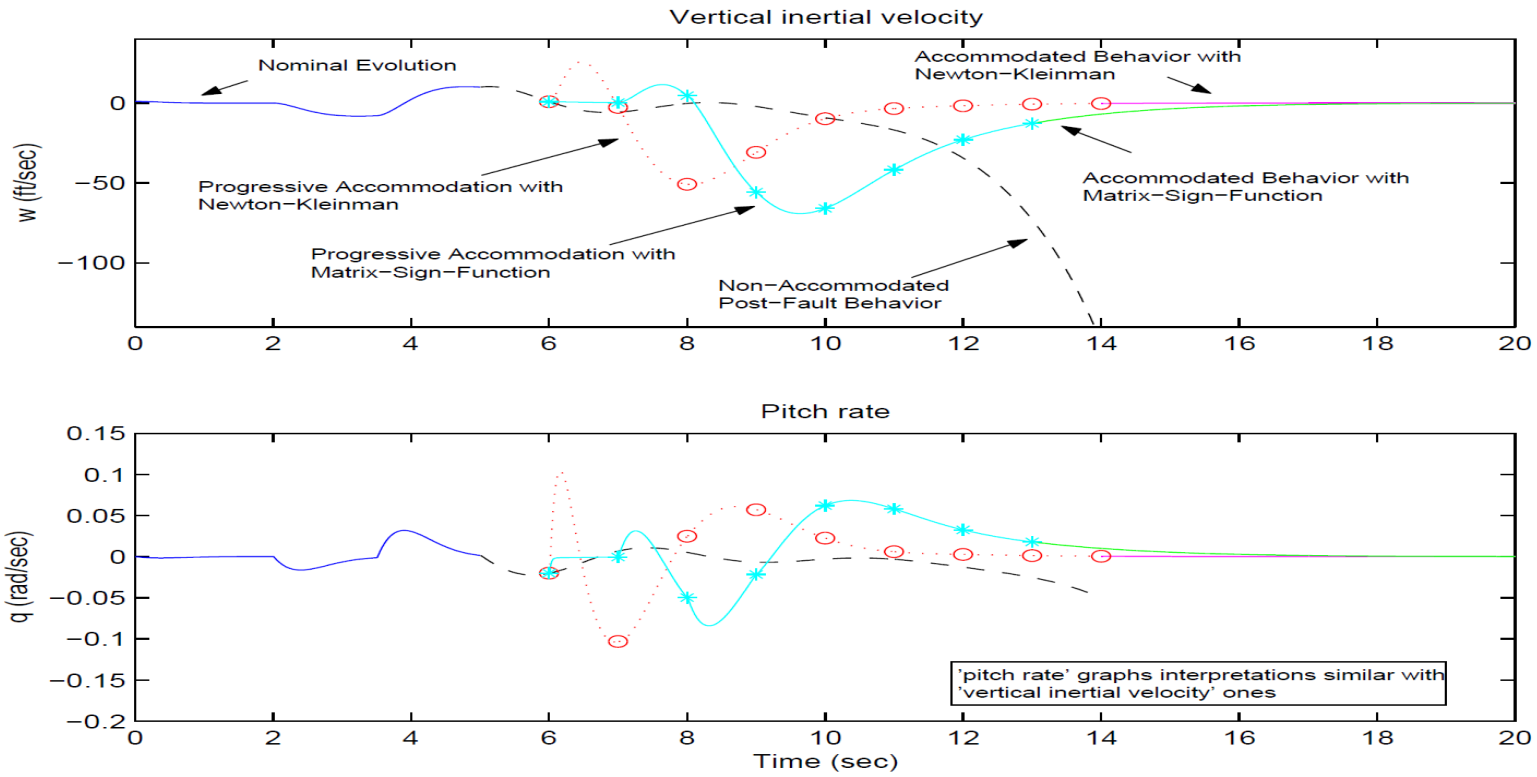
# Classical and Progressive Accommodation IX




Closed-Loop Behavior for Longitudinal B747-PH

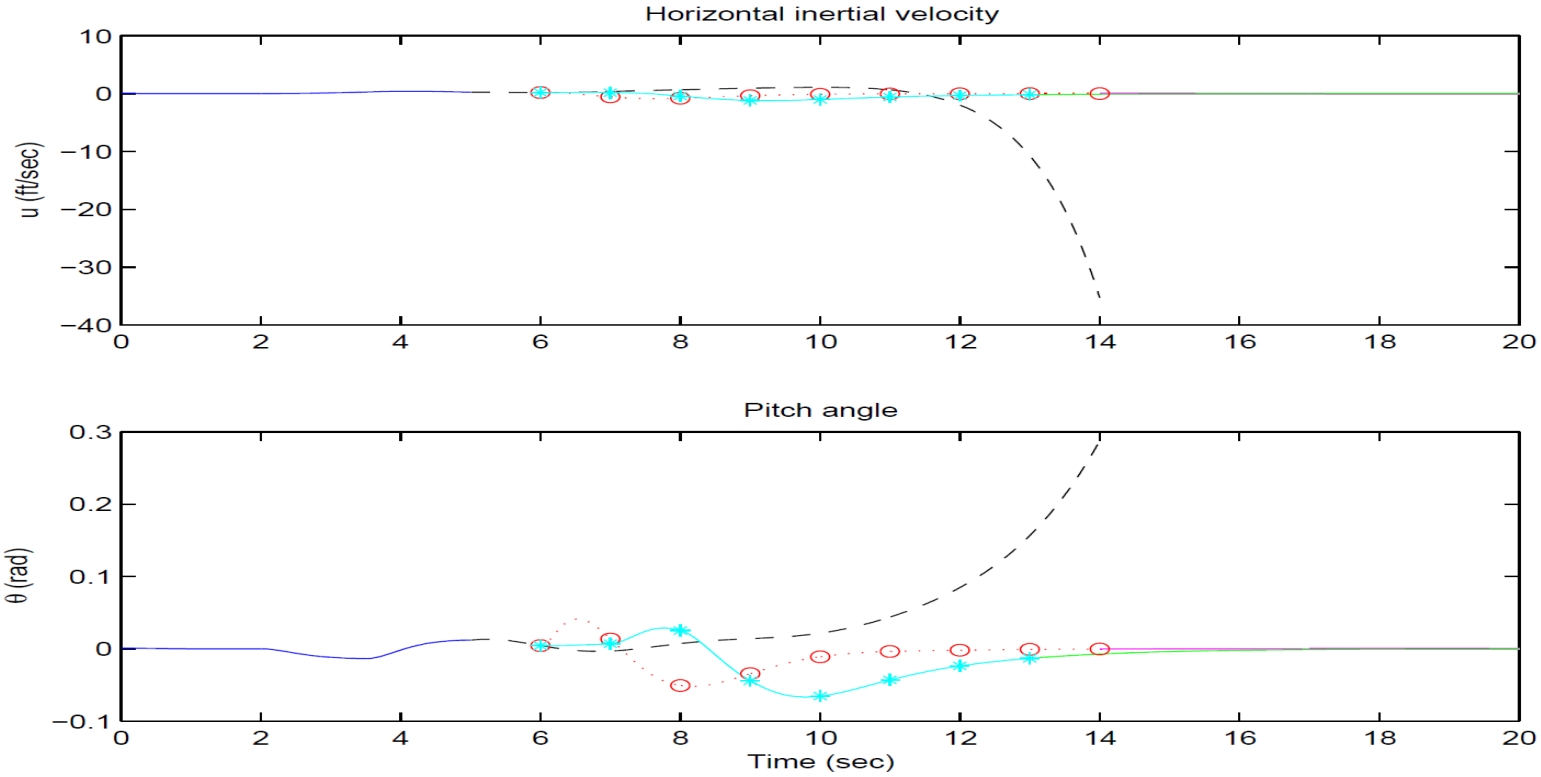
# Classical and Progressive Accommodation X

Furthermore, the post-fault progressive accommodation graphs are shown in the following figures, respectively for the short-period mode and for the phugoid mode.




Progressive Accommodation for Longitudinal B747-SP

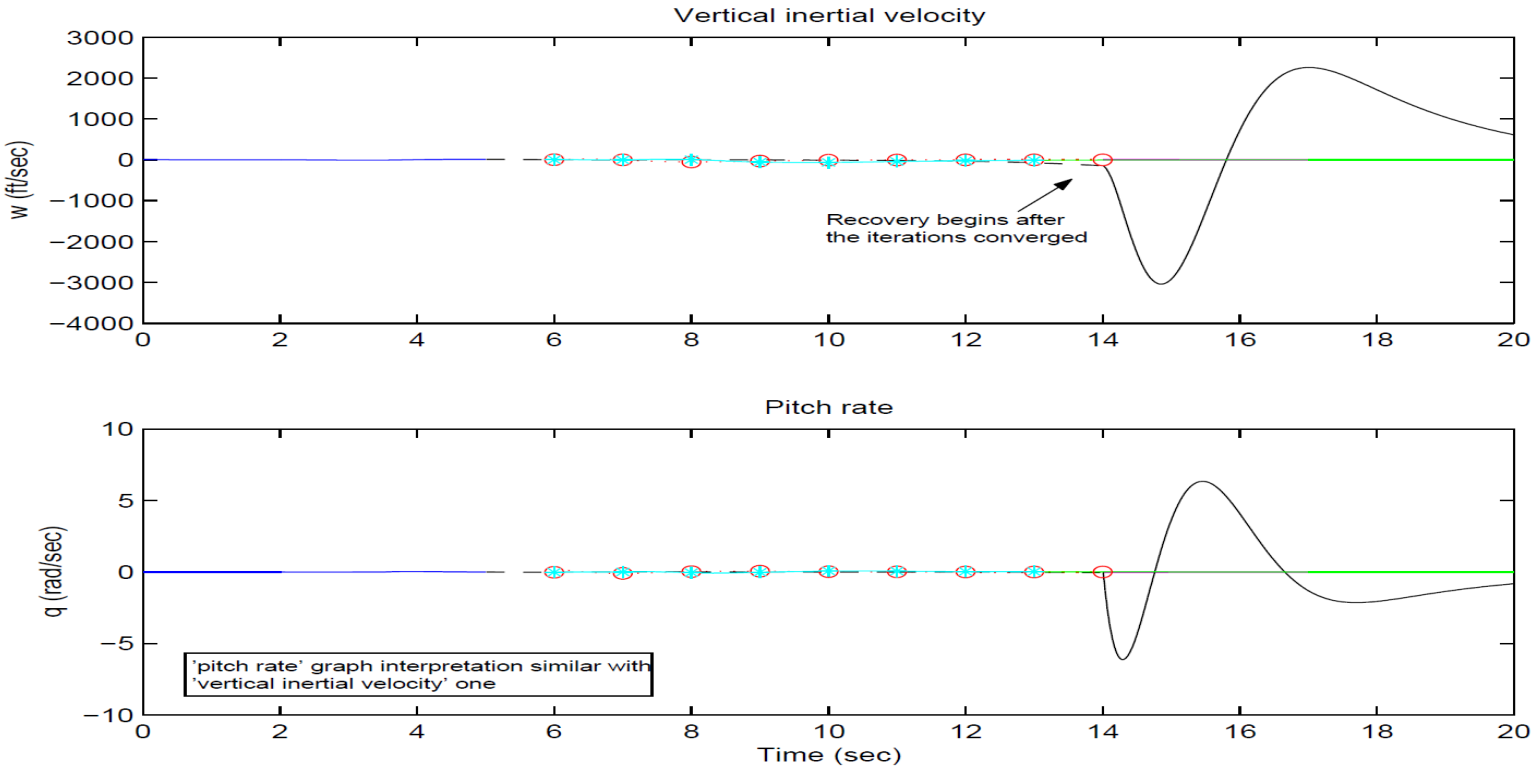
# Classical and Progressive Accommodation XI




Progressive Accommodation for Longitudinal B747-PH

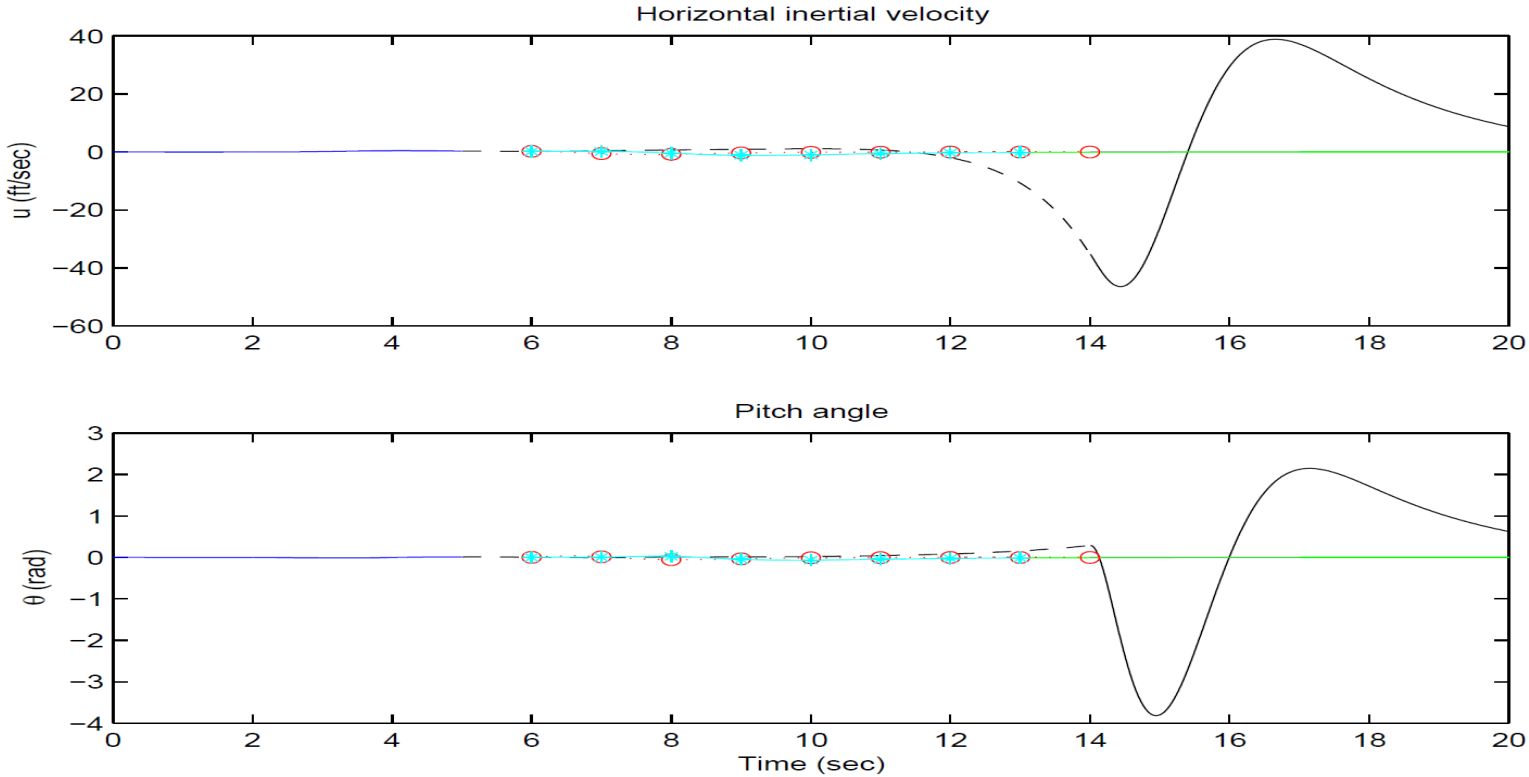
# Classical and Progressive Accommodation XII

More again, the post-fault classical accommodation graphs are shown in the following figures, respectively for the short-period mode and for the phugoid mode.




Classical Accommodation for Longitudinal B747-SP

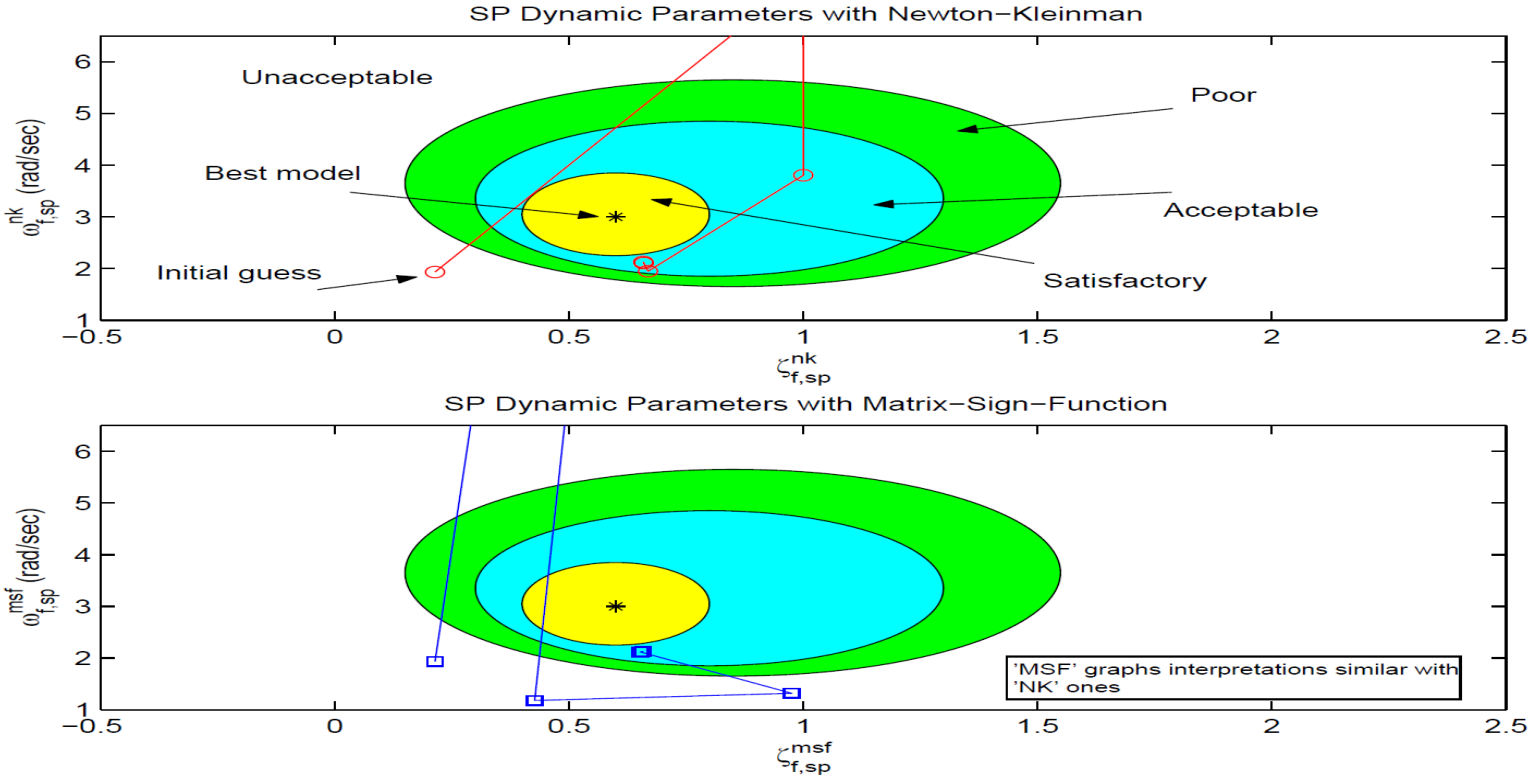
# Classical and Progressive Accommodation XIII




Classical Accommodation for Longitudinal B747-PH

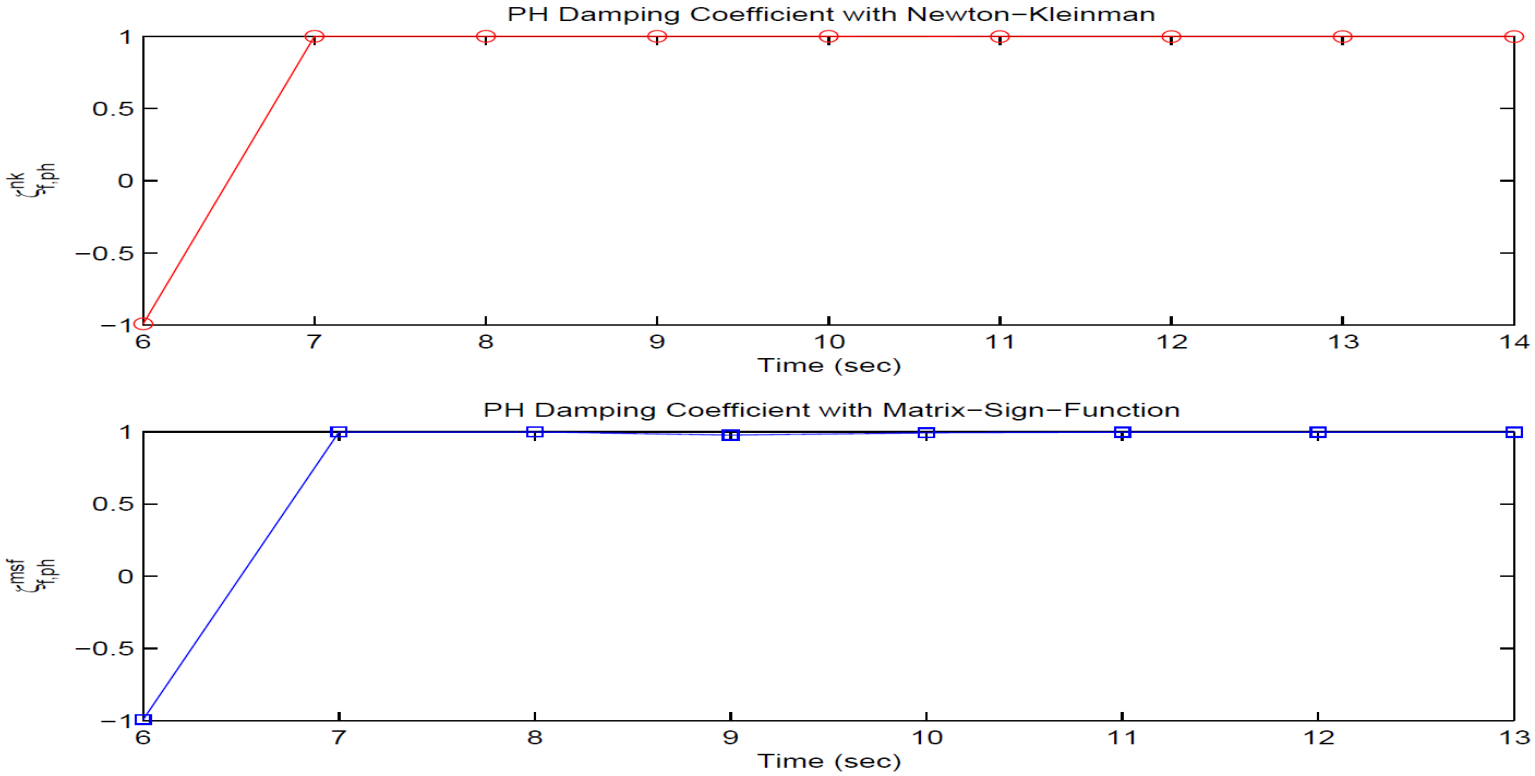
# Classical and Progressive Accommodation XIV

Still, the following figures show the evolution of the handling criteria for the longitudinal dynamics during the convergence of the iterative accommodation process, respectively for the damping coefficient and natural frequency of the short-period mode and for the damping coefficient of the phugoid mode.




Iterative Evolution for the B747-SP Longitudinal Dynamics

# Classical and Progressive Accommodation XV




Iterative Evolution for the B747-PH Longitudinal Dynamics

# Outline


1. B747 Progressive Fault Accommodation
2. B747 Admissible and Optimal Model Matching
   - Admissible Models
   - Faulty System

# Admissible Models I

It can be checked that the short-period mode provides the "best" performance as long as the damping coefficient is about $0.6$ and the natural frequency approximates at $3$ $rad/sec$; there are no specific values for the dynamical parameters of the phugoid mode but only the damping coefficient to remain positive.

Let then $(\zeta_f, \omega_f)$ be the damping coefficient and the natural frequency associated with the accommodated $2$-nd order model $\hat{M}_f$ with the closed-loop matrix

$$\hat{A}_f = A_f - B_f K_f \triangleq \begin{bmatrix} \hat{a}_{f_{11}} & \hat{a}_{f_{12}} \\ \hat{a}_{f_{21}} & \hat{a}_{f_{22}} \end{bmatrix},$$

and let there be its spectrum

$$\Lambda(\hat{A}_f) = \{\hat{\lambda}_{f_1}, \hat{\lambda}_{f_2}\};$$

# Admissible Models II

the damping coefficient and the natural frequency can then be expressed as

$$\zeta_f = -\frac{\hat{\lambda}_{f_1} + \hat{\lambda}_{f_2}}{2\sqrt{\hat{\lambda}_{f_1}\hat{\lambda}_{f_2}}} \text{ and } \omega_f = \sqrt{\hat{\lambda}_{f_1}\hat{\lambda}_{f_2}}\,,$$

that can also be developed as

$$\zeta_f = -\frac{\mathrm{tr}\,(\hat{A}_f)}{2\sqrt{\det\,(\hat{A}_f)}} = -\frac{\hat{a}_{f_{11}} + \hat{a}_{f_{22}}}{2\sqrt{\hat{a}_{f_{11}}\hat{a}_{f_{22}} - \hat{a}_{f_{12}}\hat{a}_{f_{21}}}} \text{ and}$$

$$\omega_f = \sqrt{\det\,(\hat{A}_f)} = \sqrt{\hat{a}_{f_{11}}\hat{a}_{f_{22}} - \hat{a}_{f_{12}}\hat{a}_{f_{21}}}\,, \tag{1}$$

and conditions on the closed-loop post-fault accommodated model may therefore be imposed function of the restrictive areas of the dynamical parameters from (1).

# Admissible Models III

In this respect, define the matrix inequalities of admissible behaviors for the second-order post-fault closed-loop models associated with the full longitudinal model, respectively $\hat{M}_f^{SP}$ for the short-period mode and $\hat{M}_f^{PH}$ for the phugoid mode, precisely the sets

$$\Psi_1(\hat{M}_f^{SP}) = \frac{(\zeta_f^{SP} - 0.6)^2}{0.04} + \frac{(\omega_f^{SP} - 3)^2}{0.64} - 1 \leq 0\,, \tag{2a}$$

$$\Psi_2(\hat{M}_f^{SP}) = \frac{(\zeta_f^{SP} - 0.75)^2}{0.20} + \frac{(\omega_f^{SP} - 3.45)^2}{2.25} - 1 \leq 0\,, \tag{2b}$$

$$\Psi_3(\hat{M}_f^{SP}) = \frac{(\zeta_f^{SP} - 0.85)^2}{0.45} + \frac{(\omega_f^{SP} - 3.75)^2}{4.00} - 1 \leq 0\,, \tag{2c}$$

$$\Psi_4(\hat{M}_f^{PH}) = -\zeta_f^{PH} \leq 0\,; \tag{2d}$$

hence, conditions (2a), (2b), (2c) define all short-period closed-loops whose dynamic performance are respectively "satisfactory" if all hold, "acceptable" if only conditions (2b) and (2c) hold, "poor" if only condition (2c) holds, or "unacceptable" if none among (2a), (2b), (2c) holds; note also that (2d) turns to be linear and states if the control law provides an acceptable phugoid mode behavior.

# Reference Model I

Although the B747 natural modes are inherently stable, the handling qualities of the system may be enhanced by choosing an appropriate closed-loop reference model and using an exact model matching controller design.

It can then be checked that the reference model

$$\hat{A}_n^* = \begin{bmatrix} -0.0069 & 0.0139 & -0.0006 & -32.2006 \\ 0.0489 & -0.3993 & 720.6473 & -54.0960 \\ 0.0090 & -0.0064 & -3.8468 & -3.4675 \\ 0 & 0 & 1.0000 & 0 \end{bmatrix}, \tag{3}$$

with eigenvalues

$$\Lambda\left(\hat{A}_n^*\right) = \left\{\hat{\lambda}_{n_1}^*, \hat{\lambda}_{n_2}^*\right\} = \{-2.0578 \pm 2.2020\jmath, -0.0687 \pm 0.0907\jmath\},$$

provides satisfactory performance; indeed, the reference model (3) has a damping coefficient of about $0.6$ $(0.6828)$ and a natural frequency of approximately $3$ $(3.0139)$ *rad/sec* for the short-period mode, respectively a damping coefficient of about $0.6$ $(0.6038)$ and a natural frequency of approximately $0.1$ $(0.1138)$ *rad/sec* for the phugoid mode, numerical values which the classical "thumb-print" criterion recommends as a good target for the aircraft as far as full longitudinal motion is concerned.

# Reference Model II

The state-feedback controller with

$$K_n = \begin{bmatrix} 0.0077 & -0.0047 & -2.9527 & -2.9952 \end{bmatrix}, \tag{4}$$

achieves exact model matching and it is therefore chosen as the nominal system controller.

# Considered Faults I

The considered fault is the loss-of-efficiency of the actuator that controls the elevator deflection, classically modeled by a multiplicative factor $\tau_f \in (0, 1]$, so that the linear state-space representation of the faulty system will be described by matrices $A_f = A_n$ and $B_f = (1 - \tau_f)B_n$.

# Considered Faults II

Numerical values of $K_f$ are given in the following table for increasing values of $\tau_f$. (B747-L stands for the B747 longitudinal dynamics.)

| $\tau_f$ | 0.1 | 0.2 | 0.3 | 0.4 |
|---|---|---|---|---|
| $k_{f_1}$ | 0.0086 | 0.0096 | 0.0110 | 0.0129 |
| $k_{f_2}$ | −0.0052 | −0.0059 | −0.0067 | −0.0078 |
| $k_{f_3}$ | −3.2808 | −3.6909 | −4.2182 | −4.9212 |
| $k_{f_4}$ | −3.3280 | −3.7440 | −4.2788 | −4.9920 |
| $\zeta_f^{SP}$ | 0.7122 | 0.7474 | 0.7905 | 0.8446 |
| $\omega_f^{SP}$ | 3.1617 | 3.3372 | 3.5502 | 3.8157 |
| $\zeta_f^{PH}$ | 0.6075 | 0.6113 | 0.6151 | 0.6190 |
| $\omega_f^{PH}$ | 0.1141 | 0.1145 | 0.1149 | 0.1152 |

| $\tau_f$ | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|---|---|---|---|---|---|
| $k_{f_1}$ | 0.0154 | 0.0193 | 0.0257 | 0.0386 | 0.0771 |
| $k_{f_2}$ | −0.0094 | −0.0117 | −0.0156 | −0.0234 | −0.0469 |
| $k_{f_3}$ | −5.9055 | −7.3818 | −9.8424 | −14.7637 | −29.5273 |
| $k_{f_4}$ | −5.9904 | −7.4880 | −9.9839 | −14.9759 | −29.9518 |
| $\zeta_f^{SP}$ | 0.9151 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| $\omega_f^{SP}$ | 4.1591 | 5.3921 | 9.2132 | 15.3913 | 33.1235 |
| $\zeta_f^{PH}$ | 0.6229 | 0.6269 | 0.6310 | 0.6351 | 0.6393 |
| $\omega_f^{PH}$ | 0.1156 | 0.1160 | 0.1164 | 0.1168 | 0.1172 |

# Pseudo-Inverse Method I

However, the convenient case of $A_f = A_n$ is far from being real.

In order to consider more complex situations, it is assumed that the fault is described by two coefficients, i.e., not only the control matrix $B_f = (1 - \tau_f)B_n$ but also the system matrix $A_f = (1 - \gamma_f)A_n$, where $\gamma_f \in (0, 0.1]$ is a given factor. When the compatibility conditions are no longer satisfied, Pseudo-Inverse Method (PIM) looks for an approximate solution.

# Pseudo-Inverse Method II

The following table displays the pairs (damping coefficient, natural frequency) that are obtained for $\tau_f = 0.9$ and different values of $\gamma_f$, whereas it is easily seen that the behavior is very quickly degraded as $\gamma_f$ increases.

| $\gamma_f$ | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 |
|---|---|---|---|---|---|
| $k_{f_1}^{PIM}$ | 0.0767 | 0.0762 | 0.0757 | 0.0752 | 0.0747 |
| $k_{f_2}^{PIM}$ | −0.0486 | −0.0503 | −0.0521 | −0.0538 | −0.0555 |
| $k_{f_3}^{PIM}$ | −25.2597 | −20.9920 | −16.7244 | −12.4567 | −8.1890 |
| $k_{f_4}^{PIM}$ | −29.9518 | −29.9518 | −29.9518 | −29.9518 | −29.9519 |
| $\zeta_{f_{SP}}^{PIM}$ | 0.6012 | 0.5195 | 0.4375 | 0.3554 | 0.2730 |
| $\omega_{f_{SP}}^{PIM}$ | 3.0097 | 3.0053 | 3.0006 | 2.9956 | 2.9902 |
| $\zeta_{f_{PH}}^{PIM}$ | 0.6039 | 0.6041 | 0.6042 | 0.6042 | 0.6043 |
| $\omega_{f_{PH}}^{PIM}$ | 0.1122 | 0.1107 | 0.1092 | 0.1077 | 0.1062 |

# Pseudo-Inverse Method III

| $\gamma_f$ | 0.06 | 0.07 | 0.08 | 0.09 | 0.10 |
|---|---|---|---|---|---|
| $k_{f_1}^{PIM}$ | 0.0742 | 0.0737 | 0.0732 | 0.0727 | 0.0722 |
| $k_{f_2}^{PIM}$ | −0.0573 | −0.0590 | −0.0607 | −0.0625 | −0.0642 |
| $k_{f_3}^{PIM}$ | −3.9214 | 0.3463 | 4.6139 | 8.8816 | 13.1493 |
| $k_{f_4}^{PIM}$ | −29.9519 | −29.9519 | −29.9519 | −29.9519 | −29.9519 |
| $\zeta_{f_{SP}}^{PIM}$ | 0.1903 | 0.1073 | 0.0240 | −0.0597 | −0.1438 |
| $\omega_{f_{SP}}^{PIM}$ | 2.9846 | 2.9787 | 2.9725 | 2.9660 | 2.9592 |
| $\zeta_{f_{PH}}^{PIM}$ | 0.6043 | 0.6043 | 0.6043 | 0.6043 | 0.6043 |
| $\omega_{f_{PH}}^{PIM}$ | 0.1047 | 0.1033 | 0.1018 | 0.1004 | 0.0989 |

# Pseudo-Inverse Method IV

For example, for $\tau_f = 0.9$ and $\gamma_f = 0.1$, the pair $(A_f, B_f)$ is

$$A_f = \begin{bmatrix} -0.0062 & 0.0125 & 0 & -28.9800 \\ -0.0814 & -0.2832 & 696.5789 & 0 \\ 0.0001 & -0.0009 & -0.3856 & 0 \\ 0 & 0 & 0.9000 & 0 \end{bmatrix} \text{ and } B_f = \begin{bmatrix} -0.0000 \\ -1.8061 \\ -0.1158 \\ 0 \end{bmatrix}.$$

The PIM approach provides the control-gain matrix

$$K_f^{PIM} = \begin{bmatrix} 0.0722 & -0.0642 & 13.1493 & -29.9519 \end{bmatrix},$$

which results in the closed-loop system matrix

$$\hat{A}_f^{PIM} = \begin{bmatrix} -0.0062 & 0.0125 & 0.0003 & -28.9806 \\ 0.0490 & -0.3992 & 720.3278 & -54.0961 \\ 0.0084 & -0.0083 & 1.1367 & -3.4675 \\ 0 & 0 & 0.9000 & 0 \end{bmatrix}.$$

# Pseudo-Inverse Method V

Poles being

$$\Lambda\left(\hat{A}_f^{PIM}\right) = \{0.4255 \pm 2.9284\jmath, -0.0598 \pm 0.0788\jmath\},$$

the PIM-based closed-loop is therefore unstable, with short-period dynamical parameters damping coefficient $\zeta_{f_{SP}}^{PIM} = -0.1438$ and natural frequency $\omega_{f_{SP}}^{PIM} = 2.9592$ *rad/sec*, respectively with phugoid mode dynamical parameters damping coefficient $\zeta_{f_{PH}}^{PIM} = 0.6043$ and natural frequency $\omega_{f_{PH}}^{PIM} = 0.0989$ *rad/sec*.

# Modified Pseudo-Inverse Method - Multiplicative Fault I

When the PIM-control encounters instability, addressing the stability problem is the goal of the Modified Pseudo-Inverse Method (MPIM).

The best stable closed-loop matrix obtained with MPIM is

$$\hat{A}_{f_m}^{MPIM} = \begin{bmatrix} -0.0062 & 0.0125 & 0.0000 & -28.9800 \\ -0.0509 & -0.3137 & 696.6094 & -0.0305 \\ 0.0020 & -0.0029 & -0.3836 & -0.0020 \\ 0 & 0 & 0.9000 & 0 \end{bmatrix},$$

which is associated with the feedback

$$K_{f_m}^{MPIM} = \begin{bmatrix} 0.0169 & -0.0169 & 0.0169 & -0.0169 \end{bmatrix},$$

# Modified Pseudo-Inverse Method - Multiplicative Fault II

obtained following the MPIM algorithm with $\tilde{\delta}_{f_m} = 0.0169$ from the SVD of $B_f K_{f_m}^{PIM}$ as $U_{f_m} \Sigma_{f_m} V_{f_m}^{\top}$, precisely

$$U_{f_m} = \begin{bmatrix} -0.0000 & 0.9991 & -0.0428 & 0 \\ -0.9980 & -0.0028 & -0.0639 & 0 \\ -0.0640 & 0.0427 & 0.9970 & 0 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix},$$

$$\Sigma_{f_m} = \begin{bmatrix} 59.2010 & 0 & 0 & 0 \\ 0 & 0.0000 & 0 & 0 \\ 0 & 0 & 0.0000 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$V_{f_m} = \begin{bmatrix} 0.0022 & 0.0130 & -0.9968 & -0.0789 \\ -0.0020 & -0.0002 & 0.0789 & -0.9969 \\ 0.4020 & 0.9156 & 0.0128 & -0.0000 \\ -0.9156 & 0.4020 & 0.0030 & 0.0019 \end{bmatrix},$$

with $\sigma_{f_m}^{\max} = 59.2010$.

# Modified Pseudo-Inverse Method - Multiplicative Fault III

Note that this approach indeed allows to stabilize the system but offers no real control over the behavior of the closed-loop system which results from fault accommodation, as it yields the spectrum

$$\Lambda\left(\hat{A}_{f_m}^{MPIM}\right) = \{-0.3416 \pm 1.4057\jmath, -0.0102 \pm 0.0985\jmath\},$$

which leads to a damping coefficient of about $\zeta_{f_{SP_m}}^{MPIM} = 0.2361$ and a natural frequency of approximately $\omega_{f_{SP_m}}^{MPIM} = 1.4466$ *rad/sec* for the short-period mode, that is not acceptable, respectively to a damping coefficient of about $\zeta_{f_{PH_m}}^{MPIM} = 0.1029$ and a natural frequency of approximately $\omega_{f_{PH_m}}^{MPIM} = 0.0991$ *rad/sec* for the phugoid mode.

# Modified Pseudo-Inverse Method - Multiplicative Fault IV

The following table shows the evolution of the damping coefficient and natural frequency for $\gamma_f = 0.1$ under increasing values of $\tau_f$.

| $\tau_f$ | 0.1 | 0.2 | 0.3 | 0.4 |
|---|---|---|---|---|
| $k_{f_1}^{MPIM_1}$ | 0.0080 | 0.0090 | 0.0103 | 0.0120 |
| $k_{f_2}^{MPIM_1}$ | −0.0071 | −0.0080 | −0.0092 | −0.0107 |
| $k_{f_3}^{MPIM_1}$ | 0.0169 | 0.0169 | 0.0169 | 0.0169 |
| $k_{f_4}^{MPIM_1}$ | −0.0169 | −0.0169 | −0.0169 | −0.0169 |
| $\zeta_{f_{SP}}^{MPIM_1}$ | 0.1525 | 0.1530 | 0.1534 | 0.1538 |
| $\omega_{f_{SP}}^{MPIM_1}$ | 2.4370 | 2.4367 | 2.4365 | 2.4362 |
| $\zeta_{f_{PH}}^{MPIM_1}$ | 0.1314 | 0.1310 | 0.1307 | 0.1303 |
| $\omega_{f_{PH}}^{MPIM_1}$ | 0.1141 | 0.1141 | 0.1141 | 0.1141 |

# Modified Pseudo-Inverse Method - Multiplicative Fault V

| $\tau_f$ | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|---|---|---|---|---|---|
| $k_{f_{m_1}}^{MPIM}$ | 0.0144 | 0.0169 | 0.0169 | 0.0169 | 0.0169 |
| $k_{f_{m_2}}^{MPIM}$ | −0.0128 | −0.0161 | −0.0169 | −0.0169 | −0.0169 |
| $k_{f_{m_3}}^{MPIM}$ | 0.0169 | 0.0169 | 0.0169 | 0.0169 | 0.0169 |
| $k_{f_{m_4}}^{MPIM}$ | −0.0169 | −0.0169 | −0.0169 | −0.0169 | −0.0169 |
| $\zeta_{f_{SP_m}}^{MPIM}$ | 0.1543 | 0.1550 | 0.1670 | 0.1902 | 0.2361 |
| $\omega_{f_{SP_m}}^{MPIM}$ | 2.4360 | 2.4361 | 2.1989 | 1.8609 | 1.4466 |
| $\zeta_{f_{PH_m}}^{MPIM}$ | 0.1300 | 0.1262 | 0.1210 | 0.1155 | 0.1029 |
| $\omega_{f_{PH_m}}^{MPIM}$ | 0.1141 | 0.1113 | 0.1080 | 0.1054 | 0.0991 |

# Modified Pseudo-Inverse Method - Structural Fault I

Consider again the system under actuator faults with $\tau_f = 0.5$ but also under structural impairments as $A_f = A_n + \Delta A_f \times I_n$, with $\Delta A_f = \sigma_n^{\max} \times 10^{-3}$, where $\sigma_n^{\max} = 773.9774$ is the maximum singular value of $A_n$; hence, it can be seen that MPIM provides the closed-loop control

$$K_{f_s}^{MPIM} = \begin{bmatrix} 0.0130 & -0.0130 & -0.0130 & -0.0130 \end{bmatrix},$$

from

$$K_{f_s}^{PIM} = \begin{bmatrix} 0.0152 & -0.0948 & -5.9113 & -6.0030 \end{bmatrix},$$

# Modified Pseudo-Inverse Method - Structural Fault II

and the SVD of $B_f K_{f_s}^{PIM}$ as $U_{f_s} \Sigma_{f_s} V_{f_s}^\top$, precisely

$$U_{f_s} = \begin{bmatrix} -0.0000 & 1.0000 & -0.0016 & 0 \\ -0.9980 & -0.0001 & -0.0640 & 0 \\ -0.0640 & 0.0016 & 0.9980 & 0 \\ 0 & 0 & 0 & 1.0000 \end{bmatrix},$$

$$\Sigma_{f_s} = \begin{bmatrix} 76.2422 & 0 & 0 & 0 \\ 0 & 0.0000 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$V_{f_s} = \begin{bmatrix} 0.0018 & 0.1312 & -0.9914 & 0 \\ -0.0112 & -0.8180 & -0.1083 & -0.5649 \\ -0.7016 & 0.4056 & 0.0524 & -0.5835 \\ -0.7125 & -0.3862 & -0.0524 & 0.5835 \end{bmatrix},$$

with $\sigma_{f_s}^{\max} = 76.2422$, which results in the closed-loop system matrix

$$\hat{A}_{f_s}^{MPIM} = \begin{bmatrix} 0.7671 & 0.0139 & -0.0000 & -32.2000 \\ 0.0271 & 0.3417 & 773.8591 & -0.1175 \\ 0.0076 & -0.0085 & 0.3380 & -0.0075 \\ 0 & 0 & 1.0000 & 0.7740 \end{bmatrix}.$$

# Modified Pseudo-Inverse Method - Structural Fault III

Poles being

$$\Lambda\left(\hat{A}_{f_s}^{MPIM}\right) = \{0.3512 \pm 2.5666\jmath, 0.7592 \pm 0.1199\jmath\},$$

the MPIM-based closed-loop is therefore unstable, with dynamical parameters damping coefficient $\zeta_{f_{SP_s}}^{MPIM} = -0.136$ and natural frequency $\omega_{f_{SP_s}}^{MPIM} = 2.59$ *rad/sec* for the short-period mode, respectively with dynamical parameters damping coefficient $\zeta_{f_{PH_s}}^{MPIM} = -0.988$ and natural frequency $\omega_{f_{PH_s}}^{MPIM} = 0.769$ *rad/sec* for the phugoid mode.

# Modified Pseudo-Inverse Method - Structural Fault IV

It can also be seen in the following table through the system dynamical parameters, namely the damping coefficient and natural frequency, that the system evolution remains nonstabilized under $K_{f_s}^{MPIM}$, for the structural fault introduced and under increasing values of $\tau_f$.

| $\tau_f$ | 0.1 | 0.2 | 0.3 | 0.4 |
|---|---|---|---|---|
| $\zeta_{f_{SP_s}}^{MPIM}$ | $-0.0878$ | $-0.0972$ | $-0.1081$ | $-0.1211$ |
| $\omega_{f_{SP_s}}^{MPIM}$ | 3.6039 | 3.4110 | 3.2064 | 2.9878 |
| $\zeta_{f_{PH_s}}^{MPIM}$ | $-0.9937$ | $-0.9932$ | $-0.9926$ | $-0.9919$ |
| $\omega_{f_{PH_s}}^{MPIM}$ | 0.7703 | 0.7700 | 0.7697 | 0.7692 |

| $\tau_f$ | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|---|---|---|---|---|---|
| $\zeta_{f_{SP_s}}^{MPIM}$ | $-0.1372$ | $-0.1576$ | $-0.1844$ | $-0.2241$ | $-0.2940$ |
| $\omega_{f_{SP_s}}^{MPIM}$ | 2.7519 | 2.4938 | 2.2061 | 1.8748 | 1.4708 |
| $\zeta_{f_{PH_s}}^{MPIM}$ | $-0.9909$ | $-0.9900$ | $-0.9902$ | $-0.9907$ | $-0.9918$ |
| $\omega_{f_{PH_s}}^{MPIM}$ | 0.7686 | 0.7681 | 0.7684 | 0.7690 | 0.7701 |

# Robust Optimal Model Matching - $\alpha_f = 2.00$ I

It is therefore obvious that the stabilizing gain provided by any linear quadratic regulator technique must first be found using an efficient computational technique and just afterwards to adjust it using the modified pseudo-inverse. (In the sequel, it is assumed that $\tau_f = 0.5$.)

In this respect, the post-fault closed-loop behavior approaches to that of the nominal closed-loop by minimizing the integral quadratic criterion in generalized-form $J_{f_c}(\alpha_f, \hat{A}_n^*)$, with the weighting matrices derived as follows

$$Q_{f_c} = \begin{bmatrix} 0.0000 & -0.0000 & 0.0003 & -0.0001 \\ -0.0000 & 0.0000 & -0.0024 & 0.0007 \\ 0.0003 & -0.0024 & 5.2221 & -0.3615 \\ -0.0001 & 0.0007 & -0.3615 & 0.0694 \end{bmatrix} \times 10^5 \,,$$

$$R_{f_c} = 81.8850 \,,$$

$$L_{f_c} = \begin{bmatrix} 1.2446 & -7.7591 & -484.0444 & -491.5556 \end{bmatrix} .$$

# Robust Optimal Model Matching - $\alpha_f = 2.00$ II

As the maximum absolute value of the real-part of eigenvalues of $\hat{A}_n^*$ is $2.0578$, any smaller $\alpha_f$ can be chosen to be $\alpha_f = 2.00$, and hence the accommodated post-fault closed-loop is obtained with

$$K_{f_c}^{ROMM} = \begin{bmatrix} 0.1555 & -0.0110 & 0.1611 & 8.3444 \end{bmatrix} \times 10^4 \,,$$

as follows

$$\hat{A}_{f_c}^{ROMM} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0002 \\ 0.1404 & -0.0099 & 0.1532 & 7.5354 \\ 0.0090 & -0.0006 & 0.0093 & 0.4830 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^5 \,,$$

with spectrum

$$\Lambda\left(\hat{A}_{f_c}^{ROMM}\right) = \{-2.8020, -41.7921, -6.2627 \pm 2.7610\jmath\} \,,$$

and observe that stability is recovered.

# Robust Optimal Model Matching - $\alpha_f = 2.00$ III

Not much more is obtained using the afterwards modified pseudo-inverse adjustment, precisely using the enhanced control gain matrix

$$K_{f_c}^{ROMM_a} = \begin{bmatrix} 0.1535 & -0.0108 & 0.1589 & 8.2358 \end{bmatrix} \times 10^4 \,,$$

for the difference

$$\Delta K_{f_c}^{ROMM_a} = \begin{bmatrix} -0.1555 & 0.0110 & -0.1616 & -8.3450 \end{bmatrix} \times 10^4 \,,$$

to obtain the closed-loop

$$\hat{A}_{f_c}^{ROMM_a} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0002 \\ 0.1386 & -0.0098 & 0.1513 & 7.4373 \\ 0.0089 & -0.0006 & 0.0092 & 0.4767 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^4 \,,$$

as its spectrum

$$\Lambda\left(\hat{A}_{f_c}^{ROMM_a}\right) = \{-2.8115, -41.0064, -6.2916 \pm 2.7167\jmath\}$$

does not improve significantly.

# Simplified Robust Optimal Model Matching - $\alpha_f = 2.00$ I

In this respect, the post-fault closed-loop behavior approaches to that of the nominal closed-loop by minimizing the integral quadratic criterion in generalized-form $J_{f_s}(\alpha_f, \hat{A}_n^*)$, with the weighting matrices derived as follows

$$Q_{f_s} = \begin{bmatrix} 0.0006 & -0.0001 & -0.0074 & -0.0075 \\ -0.0001 & 0.0007 & 0.0458 & 0.0466 \\ -0.0074 & 0.0458 & 2.8619 & 2.9057 \\ -0.0075 & 0.0466 & 2.9057 & 2.9514 \end{bmatrix} \times 10^3 \,,$$

$$R_{f_s} = 81.8850 \,,$$

$$L_{f_s} = \begin{bmatrix} 1.2446 & -7.7591 & -484.0444 & -491.5556 \end{bmatrix} .$$

# Simplified Robust Optimal Model Matching - $\alpha_f = 2.00$ II

Proceeding with the same smaller $\alpha_f = 2.00$, the accommodated post-fault closed-loop is obtained with

$$K_{f_s}^{ROMM} = \begin{bmatrix} 0.0282 & -0.0021 & 0.0300 & 1.5760 \end{bmatrix} \times 10^4 \,,$$

for the difference

$$\Delta K_{f_s}^{ROMM} = \begin{bmatrix} -0.0282 & 0.0021 & -0.0306 & -1.5766 \end{bmatrix} \times 10^4 \,,$$

as follows

$$\hat{A}_{f_s}^{ROMM} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0003 \\ 0.0255 & -0.0019 & 0.0349 & 1.4232 \\ 0.0016 & -0.0001 & 0.0017 & 0.0912 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^5 \,,$$

with spectrum

$$\Lambda\left(\hat{A}_{f_s}^{ROMM}\right) = \{-2.1290, -7.9521, 0.0857 \pm 7.8625\jmath\} \,,$$

and observe that stability is not recovered.

# Simplified Robust Optimal Model Matching - $\alpha_f = 2.00$ III

Nothing more is obtained using the afterwards modified pseudo-inverse adjustment, precisely using the enhanced control gain matrix

$$K_{f_s}^{ROMM_a} = \begin{bmatrix} 0.0279 & -0.0020 & 0.0296 & 1.5554 \end{bmatrix} \times 10^4 \,,$$

to obtain the closed-loop

$$\hat{A}_{f_s}^{ROMM_a} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0003 \\ 0.0251 & -0.0018 & 0.0345 & 1.4046 \\ 0.0016 & -0.0001 & 0.0017 & 0.0900 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^4 \,,$$

as its spectrum

$$\Lambda\left(\hat{A}_{f_s}^{ROMM_a}\right) = \{-2.1410, -7.8435, 0.0893 \pm 7.8442\jmath\}$$

remains unstable.

# Extended Robust Optimal Model Matching - $\alpha_f = 2.00$ I

In this respect, the post-fault closed-loop behavior approaches to that of the nominal closed-loop by minimizing the integral quadratic criterion in generalized-form $J_{f_e}(\alpha_f, \hat{A}_n^*)$, with the weighting matrices derived as follows

$$Q_{f_e} = \begin{bmatrix} 0.0006 & -0.0001 & -0.0074 & -0.0075 \\ -0.0001 & 0.0007 & 0.0458 & 0.0466 \\ -0.0074 & 0.0458 & 2.8706 & 2.9146 \\ -0.0075 & 0.0466 & 2.9146 & 2.9604 \end{bmatrix} \times 10^3 \,,$$

$$R_{f_e} = 82.8850 \,,$$

$$L_{f_e} = \begin{bmatrix} 1.2522 & -7.7638 & -486.9973 & -494.5571 \end{bmatrix} .$$

# Extended Robust Optimal Model Matching - $\alpha_f = 2.00$ II

Proceeding with the same smaller $\alpha_f = 2.00$, the accommodated post-fault closed-loop is obtained with

$$K_{f_e}^{ROMM} = \begin{bmatrix} 0.0285 & -0.0021 & 0.0303 & 1.5893 \end{bmatrix} \times 10^4 \,,$$

as follows

$$\hat{A}_{f_e}^{ROMM} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0003 \\ 0.0258 & -0.0019 & 0.0351 & 1.4352 \\ 0.0017 & -0.0001 & 0.0018 & 0.0920 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^5 \,,$$

with spectrum

$$\Lambda\left(\hat{A}_{f_e}^{ROMM}\right) = \{-2.1331, -7.9538, -0.0733 \pm 7.8839\jmath\} \,,$$

and observe that stability is recovered this time.

# Extended Robust Optimal Model Matching - $\alpha_f = 2.00$ III

Not much more is obtained using the afterwards modified pseudo-inverse adjustment, precisely using the enhanced control gain matrix

$$K_{f_e}^{ROMM_a} = \begin{bmatrix} 0.0282 & -0.0021 & 0.0299 & 1.5686 \end{bmatrix} \times 10^4 \,,$$

for the difference

$$\Delta K_{f_e}^{ROMM_a} = \begin{bmatrix} -0.0285 & 0.0021 & -0.0308 & -1.5899 \end{bmatrix} \times 10^4 \,,$$

to obtain the closed-loop

$$\hat{A}_{f_e}^{ROMM_a} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0003 \\ 0.0254 & -0.0018 & 0.0347 & 1.4165 \\ 0.0016 & -0.0001 & 0.0017 & 0.0908 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^5 \,,$$

as its spectrum

$$\Lambda\left(\hat{A}_{f_e}^{ROMM_a}\right) = \{-2.1452, -7.8432, -0.0685 \pm 7.8664\jmath\}$$

does not improve significantly.

# Classical vs. Simplified and Extended ROMM Control - $\alpha_f = 2.00$ I

From a model matching perspective, the simplified and the extended ROMM controls provide the difference from the reference model as

$$J_f\left(K_{f_s}^{ROMM_a}\right) = \|\hat{A}_{f_s}^{ROMM_a} - \hat{A}_n^*\|_F^2 = 1.9841 \times 10^{10}\,,$$
$$J_f\left(K_{f_e}^{ROMM_a}\right) = \|\hat{A}_{f_e}^{ROMM_a} - \hat{A}_n^*\|_F^2 = 2.0176 \times 10^{10}\,,$$

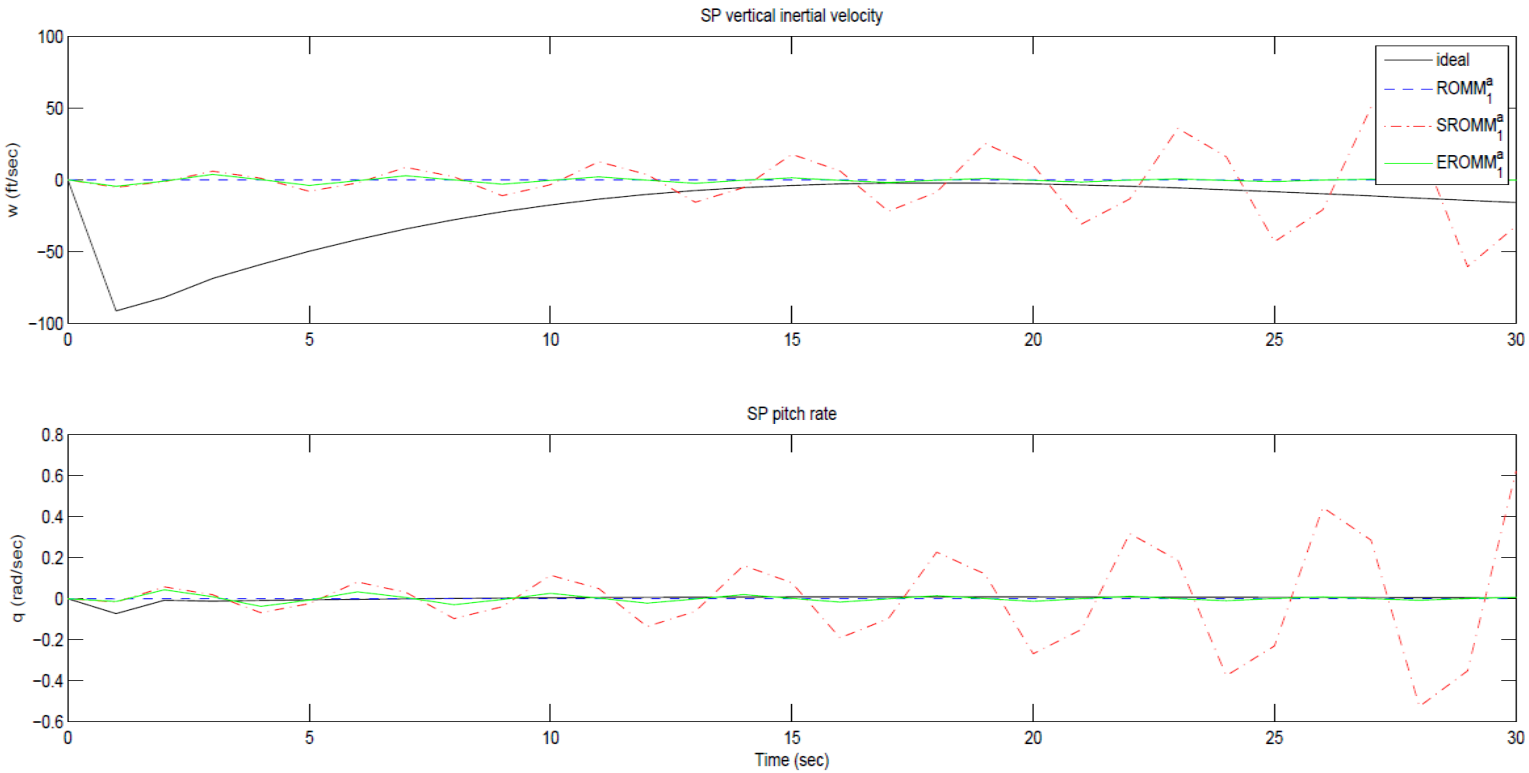
which are one order of magnitude smaller than the one evaluated for the classical ROMM-control that provides

$$J_f\left(K_{f_c}^{ROMM_a}\right) = \|\hat{A}_{f_c}^{ROMM_a} - \hat{A}_n^*\|_F^2 = 5.5590 \times 10^{11}\,,$$

and thus the simplified and the extended controls derived from the simplified and the extended ROMM approaches yield better performance than the classical control derived from the classical ROMM approach when the three approaches are used alternatively in the MPIM with correction.

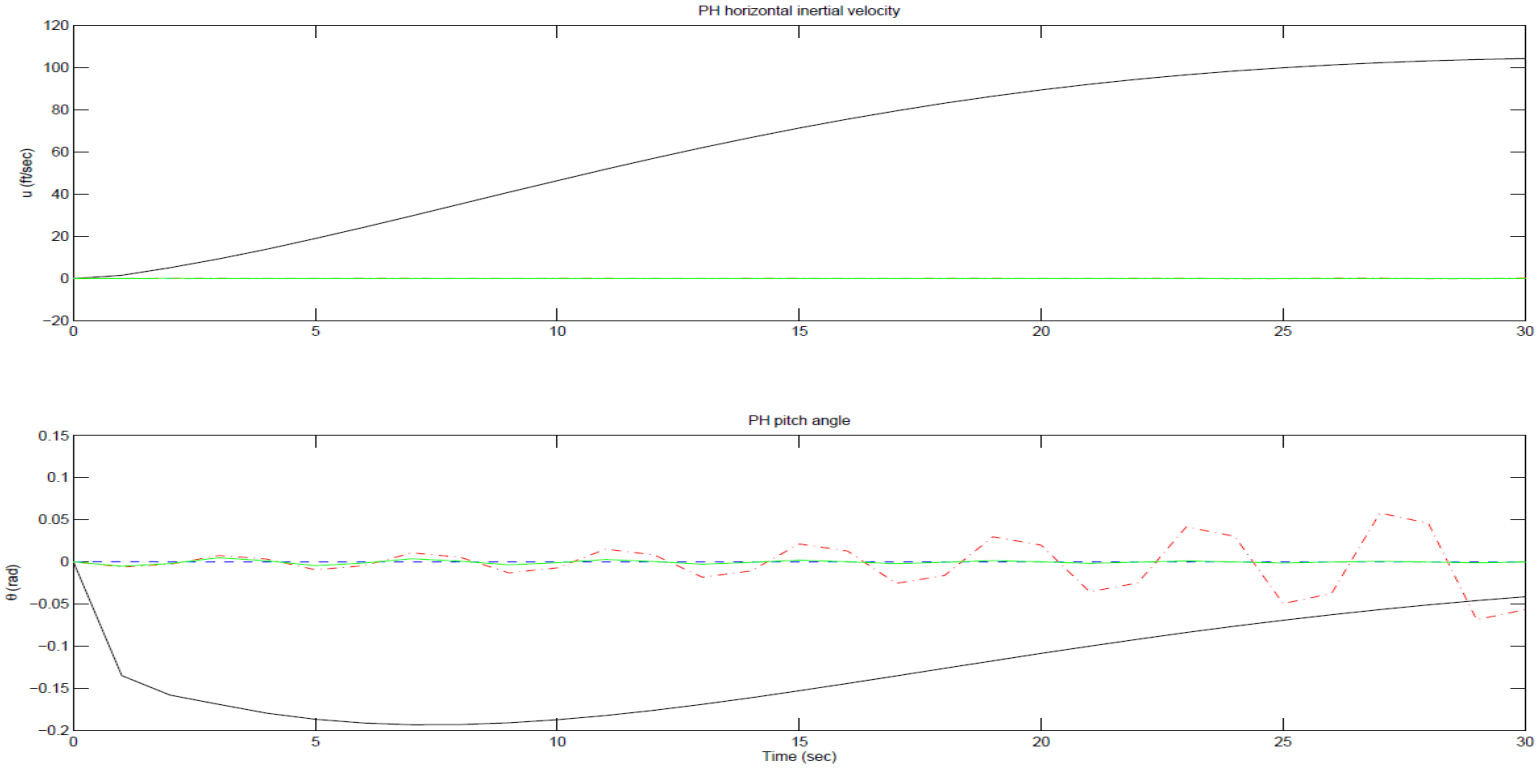
Furthermore, the simulations for the short-period mode behavior (vertical inertial velocity and pitch rate) are shown in the following first figure and those for the phugoid mode behavior (horizontal inertial velocity and pitch angle) are shown in the following second figure, respectively, during the post-fault period (restricted on the graphs to $t_f = 0$) when applying the classical respectively the simplified and the extended ROMM adjusted accommodating controls. (FA means Fault Accommodation.)

# Classical vs. Simplified and Extended ROMM Control - $\alpha_f = 2.00$ II




Optimal Accommodation with $\alpha_f = 2.00$ for B747-SP-FA

# Classical vs. Simplified and Extended ROMM Control - $\alpha_f = 2.00$ III




Optimal Accommodation with $\alpha_f = 2.00$ for B747-PH-FA

In what follows, due to instability encountered for SROMM when taking smaller $\alpha_f = 2.00$, another running example is performed for the bigger $\alpha_f = 2.50$, in spite of the rule imposing its choice.

# Robust Optimal Model Matching - $\alpha_f = 2.50$ I

In this respect, the post-fault closed-loop behavior approaches to that of the nominal closed-loop by minimizing the integral quadratic criterion in generalized-form $J_{f_c}(\alpha_f, \hat{A}_n^*)$, with the weighting matrices $Q_{f_c}, R_{f_c}, L_{f_c}$ derived as before.

As the maximum absolute value of the real-part of eigenvalues of $\hat{A}_n^*$ is $2.0578$, choose bigger $\alpha_f = 2.50$, and hence the accommodated post-fault closed-loop is obtained with

$$K_{f_c}^{ROMM} = \begin{bmatrix} 0.0255 & -0.0019 & 0.0282 & 1.4360 \end{bmatrix} \times 10^5 \,,$$

as follows

$$\hat{A}_{f_c}^{ROMM} = \begin{bmatrix} 0.0000 & -0.0000 & 0.0000 & -0.0000 \\ 0.0231 & -0.0017 & 0.0263 & 1.2968 \\ 0.0015 & -0.0001 & 0.0016 & 0.0831 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^5 \,,$$

with spectrum

$$\Lambda\left(\hat{A}_{f_c}^{ROMM}\right) = \{-3.5863, -41.8485, -7.5776 \pm 2.6691\jmath\} \,,$$

and observe that stability is recovered.

# Robust Optimal Model Matching - $\alpha_f = 2.50$ II

Not much more is obtained using the afterwards modified pseudo-inverse adjustment, precisely using the enhanced control gain matrix

$$K_{f_c}^{ROMM_a} = \begin{bmatrix} 0.0252 & -0.0019 & 0.0279 & 1.4173 \end{bmatrix} \times 10^5 \,,$$

for the difference

$$\Delta K_{f_c}^{ROMM_a} = \begin{bmatrix} -0.0255 & 0.0019 & -0.0283 & -1.4360 \end{bmatrix} \times 10^5 \,,$$

to obtain the closed-loop

$$\hat{A}_{f_c}^{ROMM_a} = \begin{bmatrix} 0.0000 & -0.0000 & 0.0000 & -0.0000 \\ 0.0228 & -0.0017 & 0.0259 & 1.2799 \\ 0.0015 & -0.0001 & 0.0016 & 0.0820 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^6 \,,$$

as its spectrum

$$\Lambda\left(\hat{A}_{f_c}^{ROMM_a}\right) = \{-3.5993, -40.9824, -7.6225 \pm 2.5917\jmath\}$$

does not improve significantly.

# Simplified Robust Optimal Model Matching - $\alpha_f = 2.50$ I

In this respect, the post-fault closed-loop behavior approaches to that of the nominal closed-loop by minimizing the integral quadratic criterion in generalized-form $J_{f_s}(\alpha_f, \hat{A}_n^*)$, with the weighting matrices $Q_{f_s}, R_{f_s}, L_{f_s}$ derived as before.

Proceeding with the same bigger $\alpha_f = 2.50$, the accommodated post-fault closed-loop is obtained with

$$K_{f_s}^{ROMM} = \begin{bmatrix} 0.0446 & -0.0033 & 0.0491 & 2.5443 \end{bmatrix} \times 10^4 \,,$$

as follows

$$\hat{A}_{f_s}^{ROMM} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0003 \\ 0.0403 & -0.0030 & 0.0521 & 2.2976 \\ 0.0026 & -0.0002 & 0.0028 & 0.1473 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^5 \,,$$

with spectrum

$$\Lambda\left(\hat{A}_{f_s}^{ROMM}\right) = \{-2.7191, -8.6767, -1.2204 \pm 8.4622\jmath\} \,,$$

and observe that surprisingly stability is recovered in this case.

# Simplified Robust Optimal Model Matching - $\alpha_f = 2.50$ II

Nothing more is obtained using the afterwards modified pseudo-inverse adjustment, precisely using the enhanced control gain matrix

$$K_{f_s}^{ROMM_a} = \begin{bmatrix} 0.0440 & -0.0033 & 0.0484 & 2.5111 \end{bmatrix} \times 10^4 \,,$$

for the difference

$$\Delta K_{f_s}^{ROMM_a} = \begin{bmatrix} -0.0446 & 0.0033 & -0.0497 & -2.5449 \end{bmatrix} \times 10^4 \,,$$

to obtain the closed-loop

$$\hat{A}_{f_s}^{ROMM_a} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0003 \\ 0.0398 & -0.0030 & 0.0515 & 2.2677 \\ 0.0025 & -0.0002 & 0.0028 & 0.1454 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^5 \,,$$

as its spectrum

$$\Lambda\left(\hat{A}_{f_s}^{ROMM_a}\right) = \{-2.7348, -8.5386, -1.2041 \pm 8.4532\jmath\}$$

remains stable.

# Extended Robust Optimal Model Matching - $\alpha_f = 2.50$ I

In this respect, the post-fault closed-loop behavior approaches to that of the nominal closed-loop by minimizing the integral quadratic criterion in generalized-form $J_{f_e}(\alpha_f, \hat{A}_n^*)$, with the weighting matrices $Q_{f_e}, R_{f_e}, L_{f_e}$ derived as before.

Proceeding with the same bigger $\alpha_f = 2.50$, the accommodated post-fault closed-loop is obtained with

$$K_{f_e}^{ROMM} = \begin{bmatrix} 0.0448 & -0.0033 & 0.0492 & 2.5523 \end{bmatrix} \times 10^4 \,,$$

as follows

$$\hat{A}_{f_e}^{ROMM} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0003 \\ 0.0405 & -0.0030 & 0.0522 & 2.3049 \\ 0.0026 & -0.0002 & 0.0029 & 0.1477 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^5 \,,$$

with spectrum

$$\Lambda\left(\hat{A}_{f_e}^{ROMM}\right) = \{-2.7246, -8.6629, -1.3161 \pm 8.4578\jmath\} \,,$$

and observe that stability is recovered.

# Extended Robust Optimal Model Matching - $\alpha_f = 2.50$ II

Not much more is obtained using the afterwards modified pseudo-inverse adjustment, precisely using the enhanced control gain matrix

$$K_{f_e}^{ROMM_a} = \begin{bmatrix} 0.0442 & -0.0033 & 0.0486 & 2.5191 \end{bmatrix} \times 10^4 \,,$$

for the difference

$$\Delta K_{f_e}^{ROMM_a} = \begin{bmatrix} -0.0448 & 0.0033 & -0.0498 & -2.5529 \end{bmatrix} \times 10^4 \,,$$

to obtain the closed-loop

$$\hat{A}_{f_e}^{ROMM_a} = \begin{bmatrix} 0.0000 & 0.0000 & 0.0000 & -0.0003 \\ 0.0399 & -0.0030 & 0.0516 & 2.2748 \\ 0.0026 & -0.0002 & 0.0028 & 0.1458 \\ 0 & 0 & 0.0000 & 0.0000 \end{bmatrix} \times 10^5 \,,$$

as its spectrum

$$\Lambda\left(\hat{A}_{f_e}^{ROMM_a}\right) = \{-2.7405, -8.5231, -1.2993 - 8.4498\jmath\}$$

does not improve significantly.

# Classical vs. Simplified and Extended ROMM Control - $\alpha_f = 2.50$ I

From a model matching perspective, the simplified and the extended ROMM controls provide the difference from the reference model as

$$J_f\left(K_{f_s}^{ROMM_a}\right) = \|\hat{A}_{f_s}^{ROMM_a} - \hat{A}_n^*\|_F^2 = 5.1696 \times 10^{10}\,,$$
$$J_f\left(K_{f_e}^{ROMM_a}\right) = \|\hat{A}_{f_e}^{ROMM_a} - \hat{A}_n^*\|_F^2 = 5.2022 \times 10^{10}\,,$$

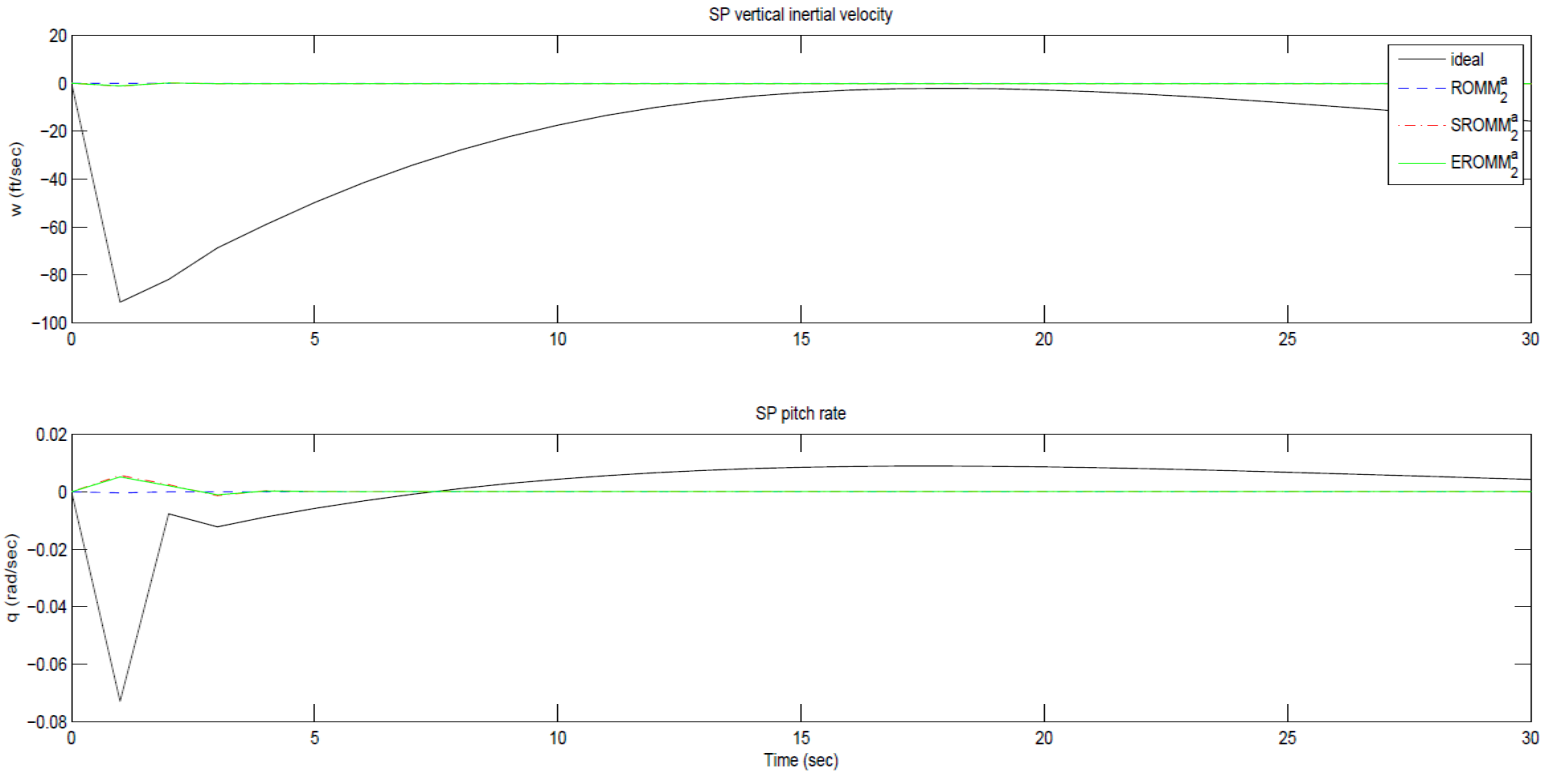
which are two orders of magnitude smaller than the one evaluated for the classical ROMM-control that provides

$$J_f\left(K_{f_c}^{ROMM_a}\right) = \|\hat{A}_{f_c}^{ROMM_a} - \hat{A}_n^*\|_F^2 = 1.6461 \times 10^{12}\,,$$

and thus similarly the simplified and the extended controls derived from the simplified and the extended ROMM approaches yield better performance than the classical control derived from the classical ROMM approach when the three are used alternatively in the MPIM with correction.

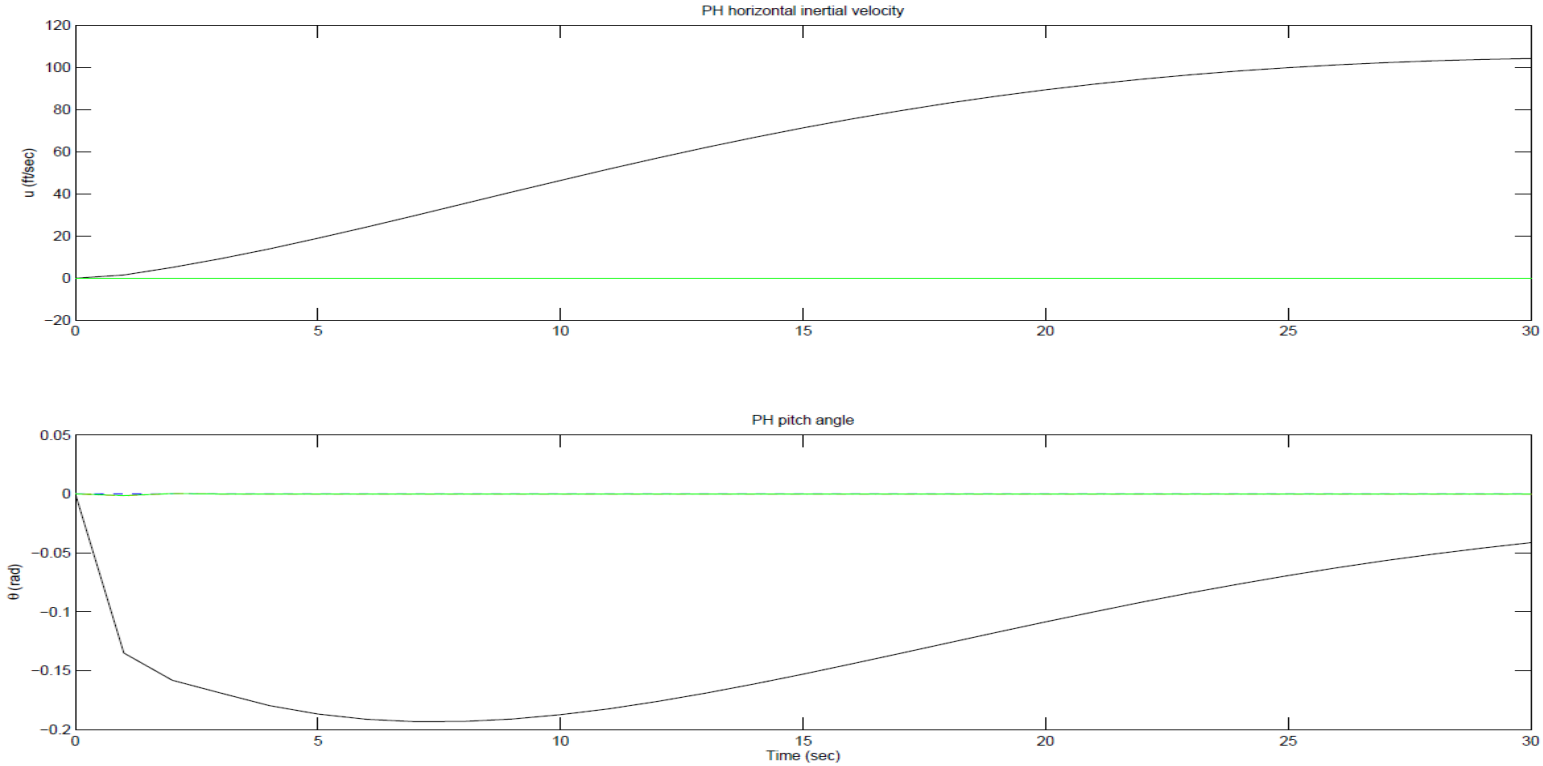
# Classical vs. Simplified and Extended ROMM Control - $\alpha_f = 2.50$ II

Furthermore, the simulations of the classical, simplified and extended ROMM fault accommodation are shown in the following four figures (focused on the part of the trajectories), precisely for the short-period mode behavior (vertical inertial velocity and pitch rate) and for the phugoid mode behavior (horizontal inertial velocity and pitch angle), respectively, during the post-fault period (restricted on the graphs to $t_f = 0$).



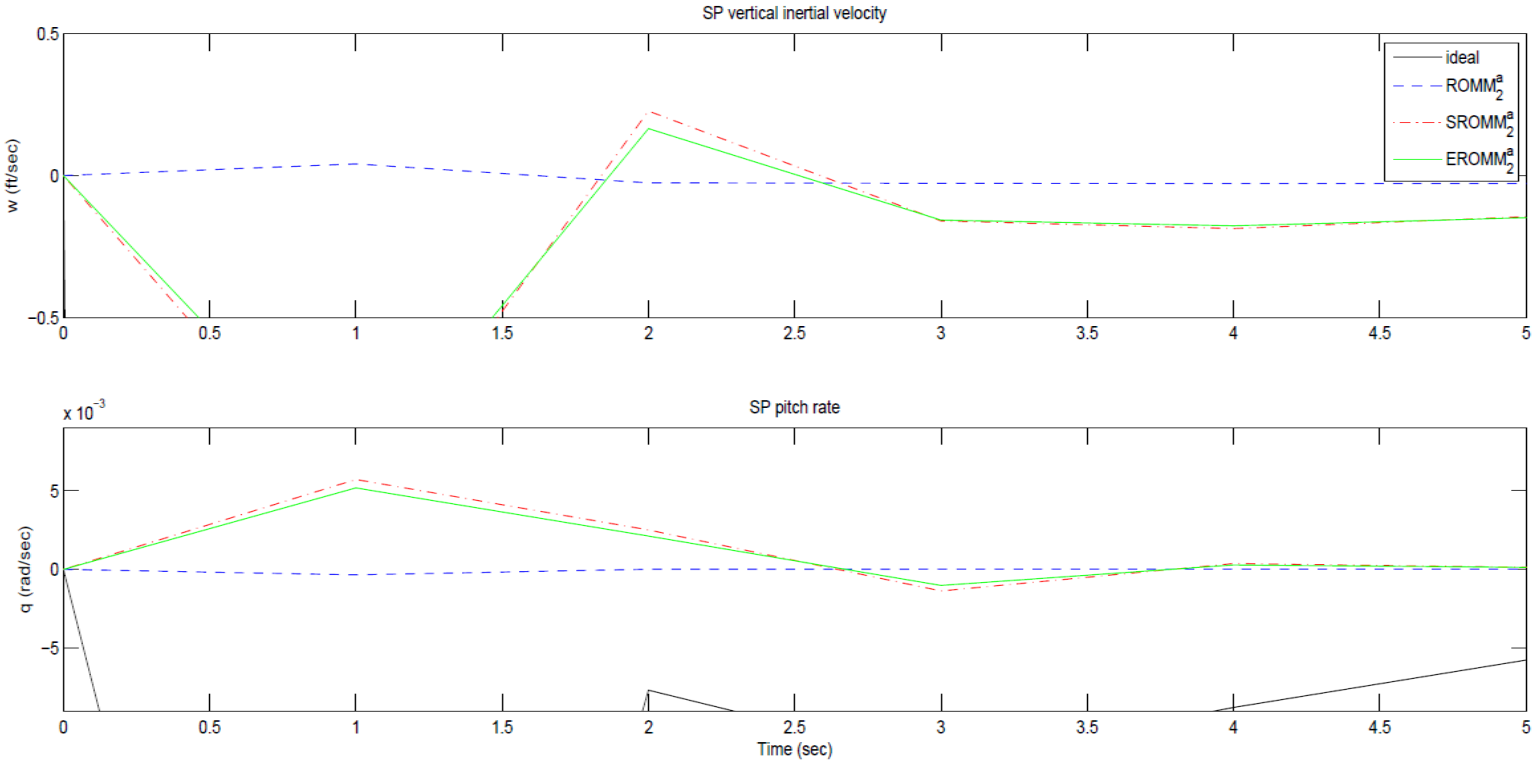

Optimal Accommodation with $\alpha_f = 2.50$ for B747-SP-FA

# Classical vs. Simplified and Extended ROMM Control - $\alpha_f = 2.50$ III



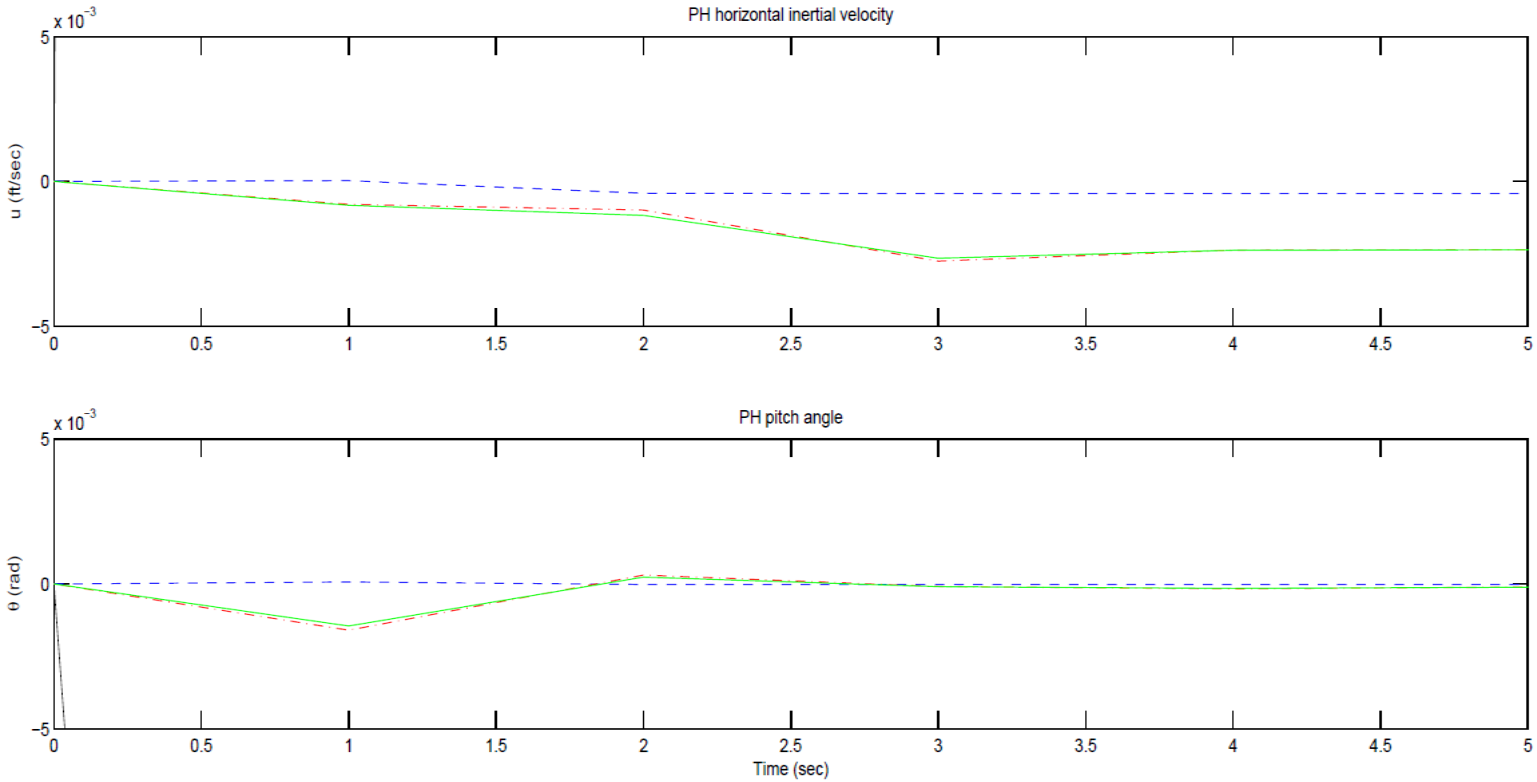

Optimal Accommodation with $\alpha_f = 2.50$ for B747-PH-FA

# Classical vs. Simplified and Extended ROMM Control - $\alpha_f = 2.50$ IV




Optimal Accommodation with $\alpha_f = 2.50$ for B747-SP-FA (Focused)

# Classical vs. Simplified and Extended ROMM Control - $\alpha_f = 2.50$ V




Optimal Accommodation with $\alpha_f = 2.50$ for B747-PH-FA (Focused)

See that in this case the graphs from the three situations follow the same shape, indicating that the bigger $\alpha_f$ is better than the smaller one, possibly compensating for the term removed from the classical robust optimal model matching criterion.

# Conclusion I

This lecture provided the full application of the fault tolerant control methods developed within these series of lectures for the longitudinal motion of the Boeing 747 aircraft.

Questions? vs. Answers!

Acknowledgments:
**Marcel Staroswiecki**, **Dumitru Popescu**,
and
**Andrei Sperilă**, **Sabin Diaconescu**, **Teodor Rotaru**,
and
Nicolai Christov, Cristian Oară.